RAJALAKSHMI ENGINEERING COLLEGE
RAJALAKSHMI NAGAR, THANDALAM – 602 105

RAJALAKSHMI ENGINEERING COLLEGE

CS23334 – Fundamentals of Data Science Lab

Laboratory Record Note Book

Name : B. SIDDHARTH

Year / Branch / Section : II / B.E. CSE H

University Register No. : 2116250701715

College Roll No. : 250701715

Semester : III

Academic Year : 2025 – 2029

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23334 – Fundamentals of Data Science Lab

NAME	SIDDHAARTH.B
ROLL NO	2560171S
YEAR	II
SEC	H

RAJALAKSHMI ENGINEERING COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: .....SIDDHARTH.B.....

Academic Year: 2025-2026 **Semester:** ...III..... **Branch:** ...CSE.....

Register No. 211625701715

Certified that this is the bonafide record of work done by the above student in the .....FUNDAMENTAL...OF..DATA..SCIENCE..... ***Laboratory during the academic year 2026 – 2027.***

2/11/26

Signature of Faculty in-charge

Submitted for the Practical Examination held on ....06/11/2026...

Internal Examiner

External Examiner

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Sl.NO	Experiment	Date	GitHub
1.a	Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.	29/07/26	
1.b	Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.	29/07/26	
1.c	Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.	05/08/26	
1.d	Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.	05/08/26	
2	Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.	12/08/26	
3.a	Conduct an experiment to show data visualization using line plot	19/08/26	
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4	To Count the frequency of occurrence of a word in a body of text is often needed during text processing.	26/08/26	
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18	Experiment to understand KNN algorithm for a given data set.	07/10/26	
19	Experiment to understand Logistic Regression for a given data set.	07/10/26	
20	Experiment to understand K-means clustering algorithm for a given data set.	07/10/26	

S.No	List Of Experiments	CO
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1.c	Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.	1
1.d	Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.	1
2	Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.	2
3.a	Conduct an experiment to show data visualization using line plot	2
3.b	Conduct an experiment to show data visualization using bar chart	2
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7	Random Sampling and Sampling Distribution	4
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RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB MANUAL

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

VISION & MISSION

OF

COMPUTER SCIENCE AND ENGINEERING DEPARTMENT

Vision

To promote highly ethical and innovative computer professionals through excellence in teaching, training and research.

Mission

- To produce globally competent professionals, motivated to learn the emerging technologies and to be innovative in solving real world problems.
- To promote research activities amongst the students and the members of faculty that could benefit the society.
- To impart moral and ethical values in their profession.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of networking, security, data mining, web technology, mobile communication and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POs)

PO1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration forthe public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and normsof the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with theengineering community and with society at large, such as, being able to comprehend and write effectivereports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one‟s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage inindependent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Computer Science and Engineering Program will demonstrate,

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areasrelated to algorithms, system software, web design, machine learning, data analytics, and networking for efficient design of computer-based systems of varying complexity. Familiarity and practicalcompetence with a broad range of programming language and open source platforms.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computationaltask, model real world problem using appropriate data structure and suitable algorithm. To understand the standard practices and strategies in software project development, using open- ended programmingenvironments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify researchgaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible computer science professional.

Objectives:
• To understand the fundamental concepts of data science.
• To understand data cleaning and processing for data science.
• To learn the fundamentals of exploratory data analysis (EDA).
• To understand inferential data analytics
• To understand the Machine Learning concepts.

Course Outcomes: On completion of course, students will be able to
• Explore the fundamental concepts of data science
• Explain how data is collected, managed and processed for data science
• Understand the fundamentals of exploratory data analysis.
• Perform Statistical Inference and Analyze Variance in Data
• Understand various machine learning algorithms used in data science process.

CO - PO – PSO matrices of course

COs/POs&PSOs	PO1	PO2	PO3	PO4	PO5	PO 6	PO7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO3
CS23334.1	2	2	1	2	2	-	-	-	1	1	1	2	2	2	2
CS23334.2	2	1	-	1	1	-	-	-	2	1	1	2	2	3	1
CS23334.3	2	2	1	2	2	1	1	-	1	2	1	3	2	2	3
CS23334.4	3	2	2	1	2	-	-	-	1	1	2	2	3	3	2
CS23334.5	2	2	1	2	2	-	-	-	1	1	1	2	2	2	2
Average	2.2	1.8	1.25	1.6	1.8	1	1	-	1.2	1.2	1.2	2.2	2.2	2.4	2

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1.d	Conduct an experiment to encrypt and decrypt given sensitive data Use the cryptography library fernet.	1
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3.a	Conduct an experiment to show data visualization using line plot	2
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:

Exp No:**1.a** Analyze the trend of data science job postings over the last decade

Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
        'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

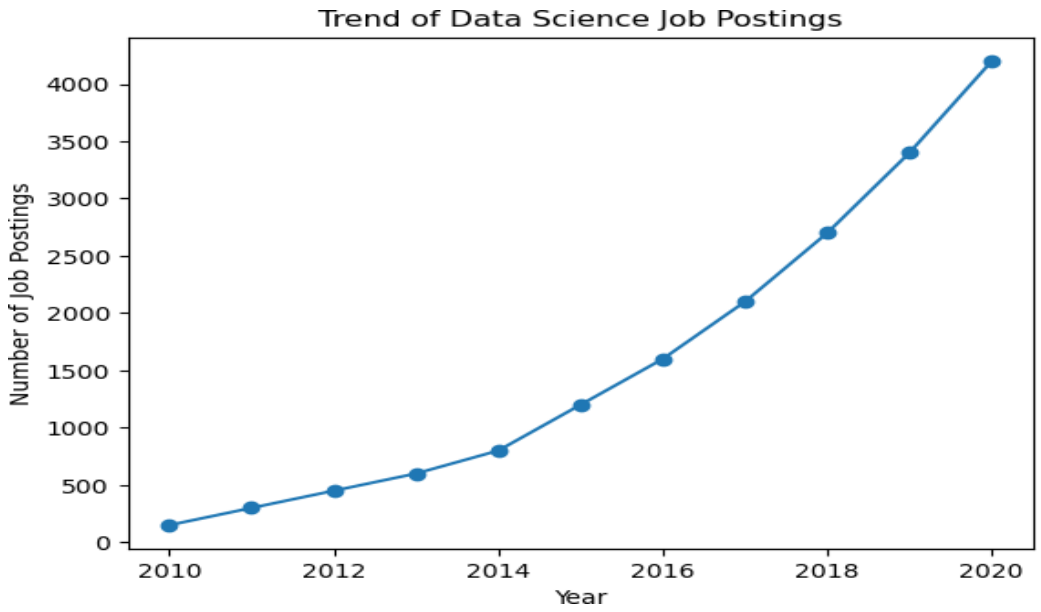
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

Sample Data Input:

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

Sample Output:



Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']

counts = [300, 500, 450, 200, 150]

plt.bar(roles, counts)

plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Count')

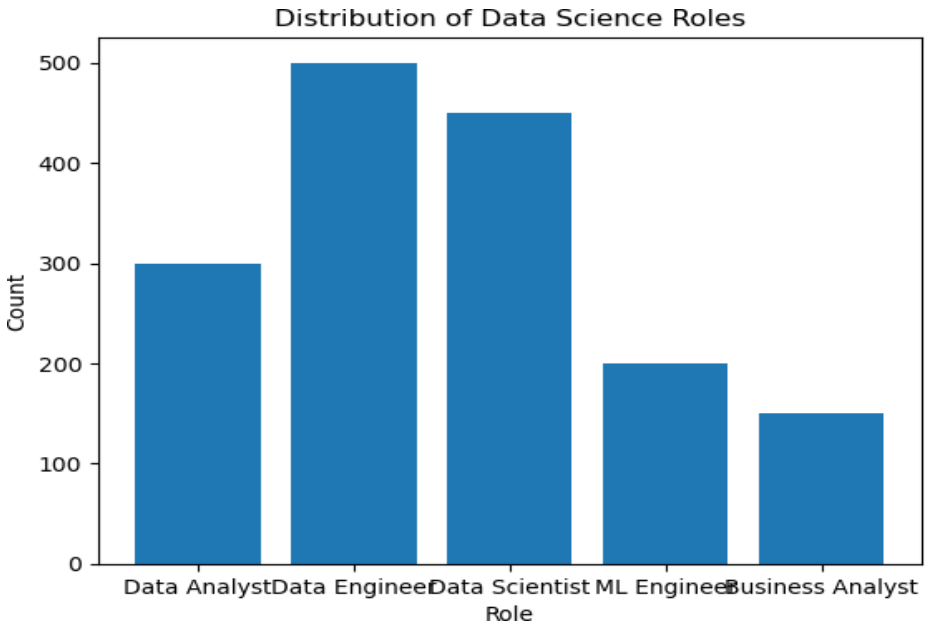
plt.show()
```

Sample Data Input:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst.

counts = 300, 500, 450, 200, 150.

Sample Output:



Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description: Create small datasets for each type and explain their characteristics.

```
# Structured data example
structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)


# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image, or a video file."
print("\nUnstructured Data:\n", unstructured_data)


# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an
image, or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}
}
```

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Description: Use the cryptography library to encrypt and decrypt a piece of data.

```
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn
XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8Fov1tV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```

Exp No:**2** Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

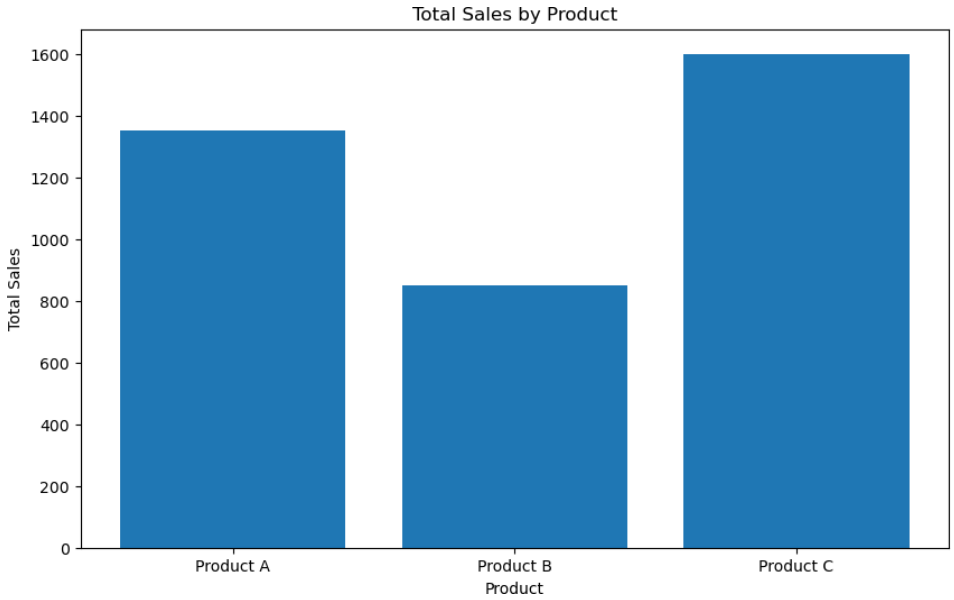
Description: Use sample data set sales-data.csv.

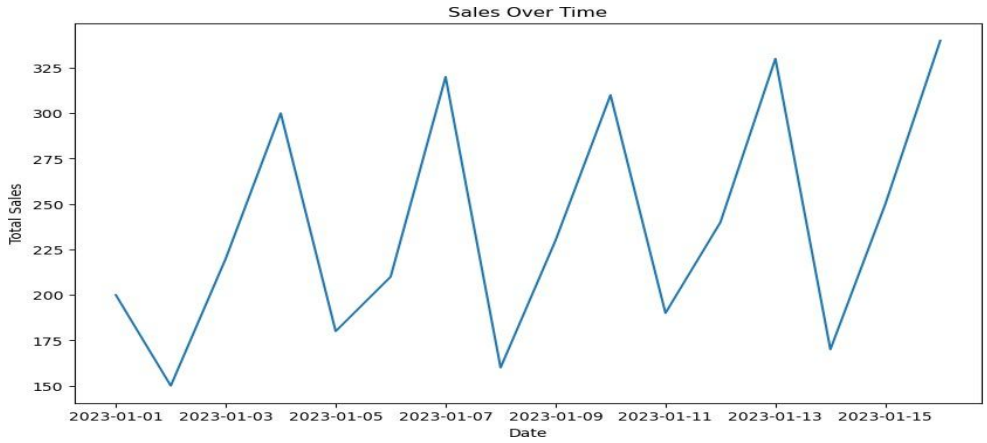
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)
# Display the first few rows of the DataFrame
print(df.head())
# Check for missing values
print(df.isnull().sum())
# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
# Summary statistics
print(df.describe())
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)
```

```
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)
# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

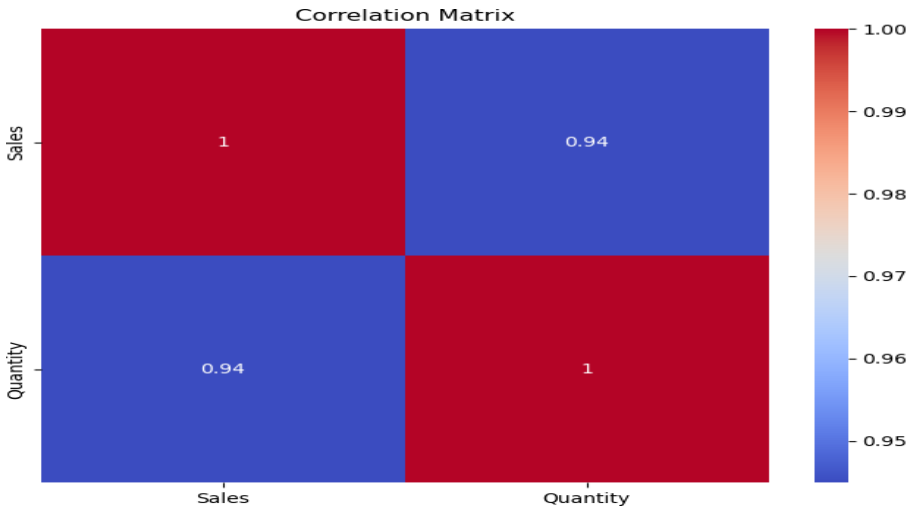
OUTPUT:

```
Date     Product  Sales  Quantity Region
0   2023-01-01  Product A    200         4  North
1   2023-01-02  Product B    150         3  South
2   2023-01-03  Product A    220         5  North
3   2023-01-04  Product C    300         6   East
4   2023-01-05  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```






```
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0
              Sales  Quantity
Sales      1.000000  0.944922
Quantity   0.944922  1.000000
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: Futur
eWarning: The default value of numeric_only in DataFrame.corr is deprecated
. In a future version, it will default to False. Select only valid columns
or specify the value of numeric_only to silence this warning.
  correlation_matrix = df.corr()
```



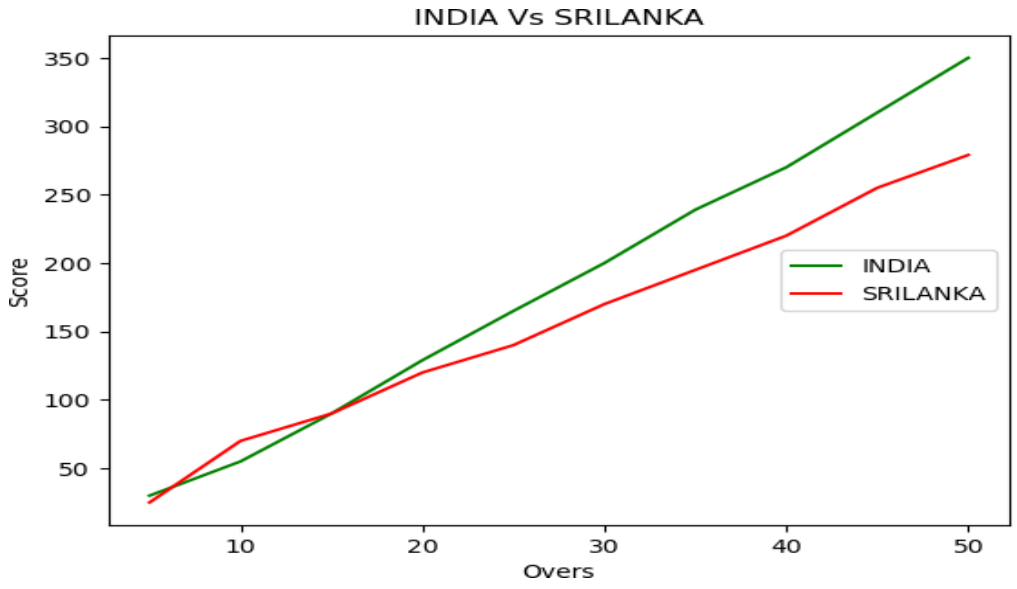
Exp No:3.a Conduct an experiment to show data visualization using line plot

.

Description: Take any sample data either through csv file or data fetched directly through code.

```
import matplotlib.pyplot as cricket
Overs=list(range(5,51,5))
Indian_Score=[30,55,90,129,165,200,239,270,310,350]
Srilankan_Score=[25,70,90,120,140,170,195,220,255,279]
cricket.plot(Overs,Indian_Score)
cricket.plot(Overs,Srilankan_Score)
cricket.show()
cricket.title("INDIA Vs SRILANKA")
cricket.xlabel("Overs")
cricket.ylabel("Score")
cricket.legend()
cricket.plot(Overs,Indian_Score,color="green",label="INDIA")
cricket.plot(Overs,Srilankan_Score,color="red",label="SRILANKA")
cricket.legend(loc="center right")
```

OUTPUT:



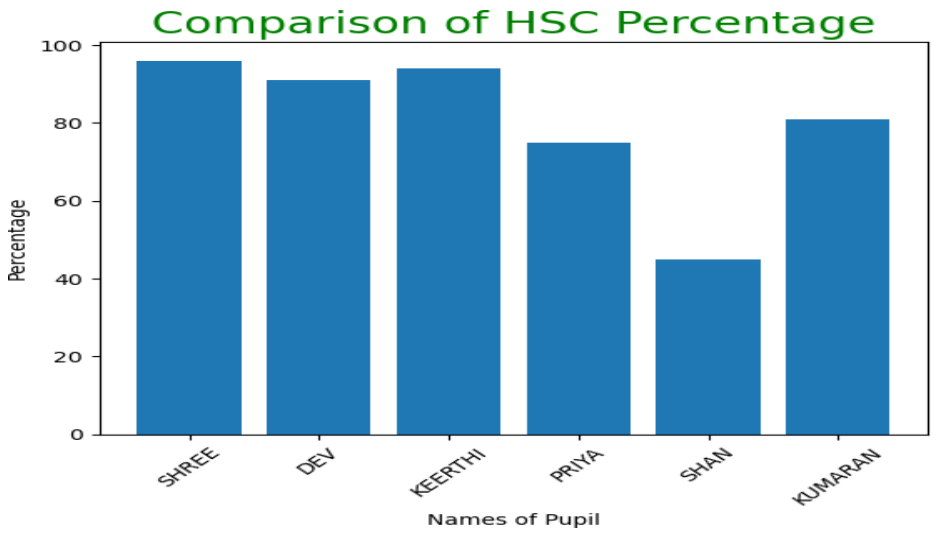
Exp No:3.b Conduct an experiment to show data visualization using bar chart

.

Description: Take any sample data either through csv file or data fetched directly through code.

```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']
xaxis = np.arange(len(Names))
Percentage_hsc = [96, 91, 94, 75, 45, 81]
hscmark.bar(Names, Percentage_hsc)
hscmark.xticks(xaxis, Names, rotation=45)
hscmark.xlabel("Names of Pupil")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")
hscmark.show()
```

OUTPUT:

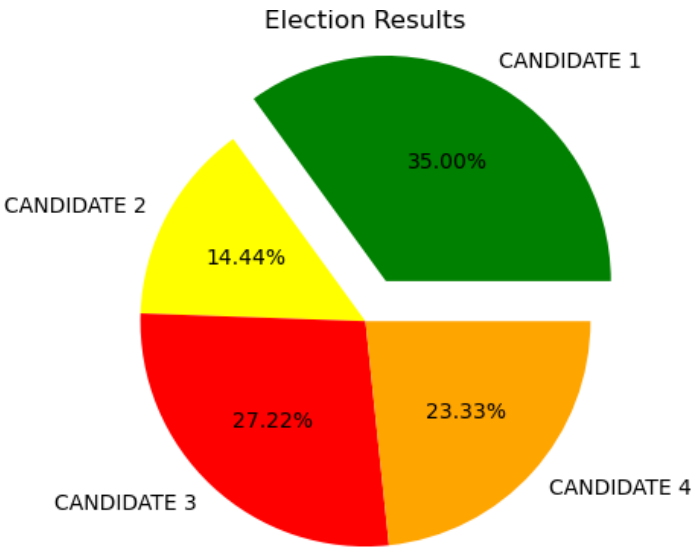


Exp No:3.c Conduct an experiment to show data visualization using pie chart

.**Description:** Take any sample data either through csv file or data fetched directly through code.

```
import matplotlib.pyplot as election
# Election data
labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']
Votes = [315, 130, 245, 210]
colors = ['green', 'yellow', 'red', 'orange']
explode = (0.2, 0, 0, 0)
# Plotting the pie chart
election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')
election.title('Election Results')
election.show()
```

OUTPUT:



Exp No:4 To Count the frequency of occurrence of a word in a body of text is often needed during text processing.

Description: Import the word_tokenize function and gutenberg.

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
# Download the gutenberg corpus if not already installed
nltk.download('gutenberg')
nltk.download('punkt')  # Ensure the punkt tokenizer models are also downloaded
# Load the text from the gutenberg corpus
sample = gutenberg.raw("austen-emma.txt")  # Change to an existing text from the corpus
# Tokenize the sample text
token = word_tokenize(sample)
# Create a list of the first 50 tokens
wlist = []
for i in range(50):
    wlist.append(token[i])
# Calculate the frequency of each word in the list
wordfreq = [wlist.count(w) for w in wlist]
# Print the word-frequency pairs
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

OUTPUT:

```
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816',
1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma
', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever'
, 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1
), ('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposi
tion', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1
), ('of', 2), ('the', 2), ('best', 1), ('blessings', 1), ('of', 2), ('e
xistence', 1), (';', 1), ('and', 3), ('had', 1), ('lived', 1), ('nearly
', 1), ('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2), ('world'
, 1), ('with', 2)]
```

Exp No:5 Data Collection and Initial Exploration

Objective: To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

- Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
- Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. **Data Collection:**

```
import pandas as pd

# Load the dataset
diabetes_df = pd.read_csv("diabetes.csv")

# Display the first few rows
print(diabetes_df.head())
```

```
[1] import pandas as pd
    db= pd.read_csv("/content/drive/MyDrive/PIMA Indian Dataset.csv")
```

```
print(db.head())
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

2. Initial Exploration:

```python
# Dataset structure and summary statistics
print(db.info())
print(db.describe())

# Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns

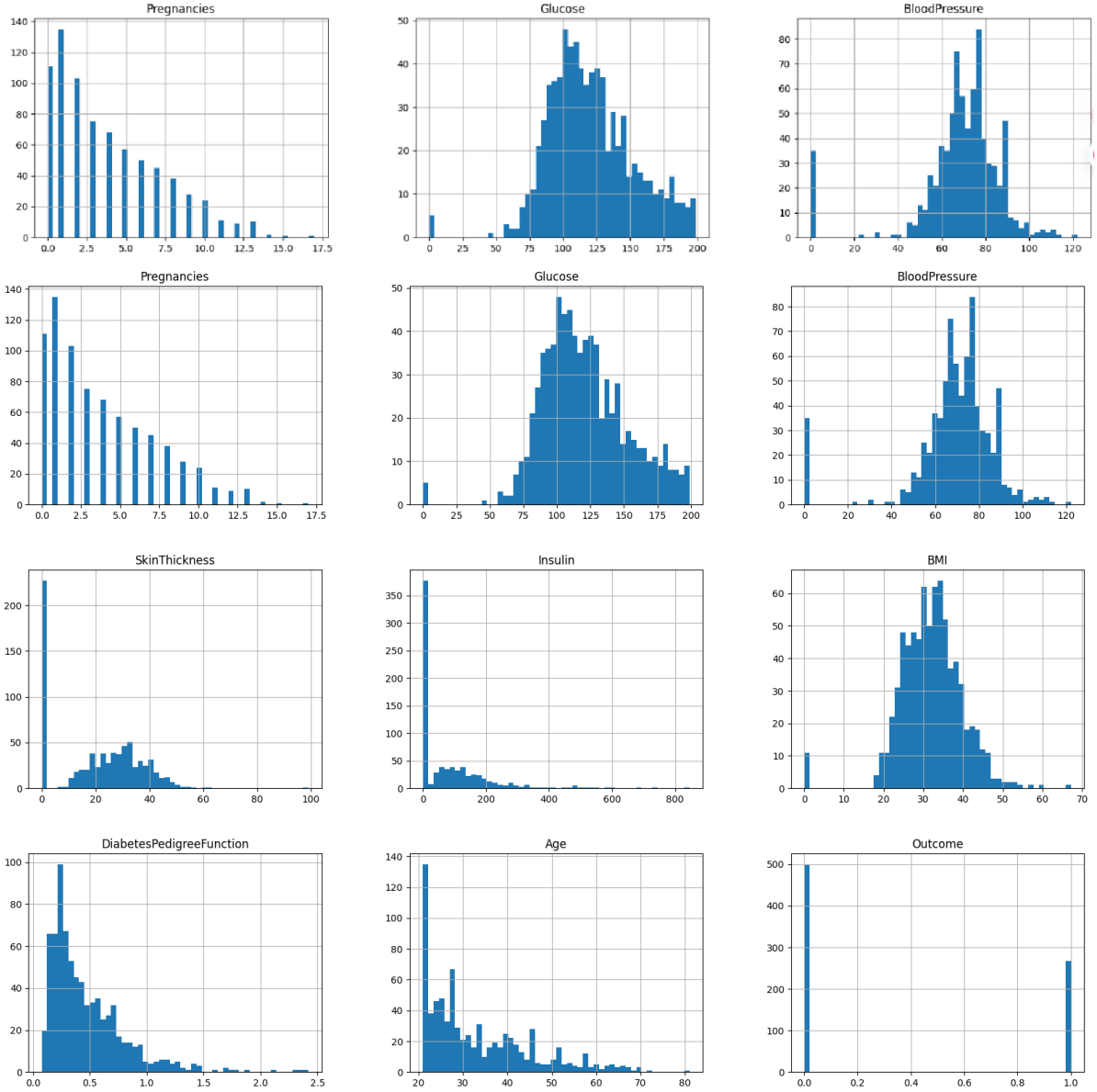
db.hist(bins=50, figsize=(20, 15))
plt.show()

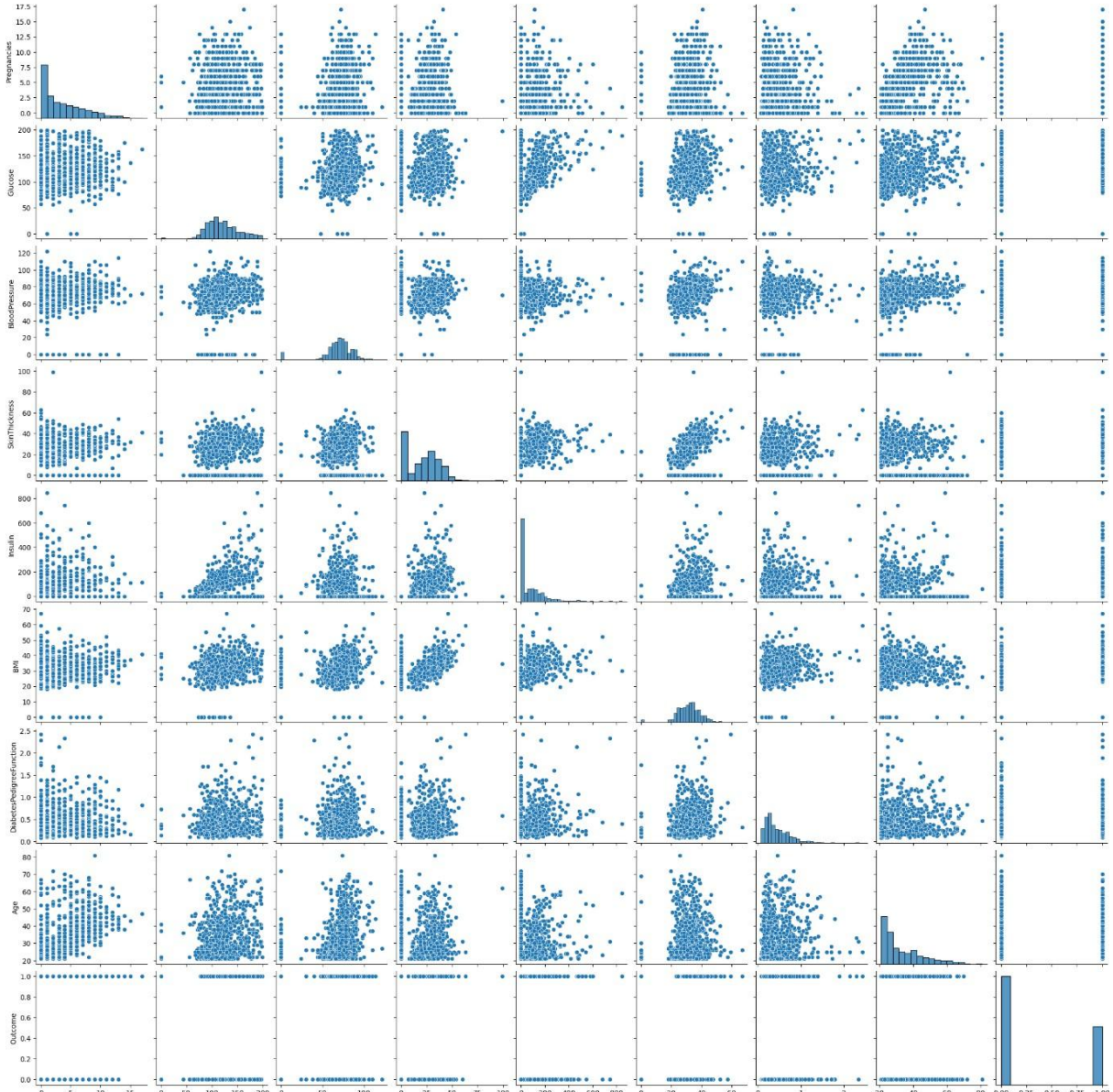
# Pairplot to visualize relationships
sns.pairplot(db)
plt.show()
```

[3]

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    int64  
 1   Glucose                   768 non-null    int64  
 2   BloodPressure             768 non-null    int64  
 3   SkinThickness             768 non-null    int64  
 4   Insulin                   768 non-null    int64  
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64  
 8   Outcome                   768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000   
mean      3.845052  120.894531      69.105469      20.536458   79.799479   
std       3.369578   31.972618      19.355807      15.952218  115.244002   
min       0.000000    0.000000       0.000000       0.000000    0.000000   
25%       1.000000   99.000000      62.000000       0.000000    0.000000   
50%       3.000000  117.000000      72.000000      23.000000   30.500000   
75%       6.000000  140.250000      80.000000      32.000000  127.250000   
max      17.000000  199.000000     122.000000      99.000000  846.000000   
```

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000




Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome

.

Experiment 6: Time Series Analysis of Website Traffic

Objective: To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Dataset: A dataset containing daily website traffic data with columns such as `Date`, `Page_Views`, `Unique_Visitors`, and `Bounce_Rate`.

Methodology:

1. **Load and Clean Data:**
    - Import the dataset and parse the `Date` column as a datetime object.
    - Handle missing values by interpolation.
2. **Time Series Decomposition:**
    - Decompose the time series data of `Page_Views` to identify trend, seasonality, and residuals.
3. **Visualization:**
    - Plot the time series data for `Page_Views`, `Unique_Visitors`, and `Bounce_Rate`.
    - Use moving averages to smooth the time series and highlight trends.
    - Create seasonal plots to visualize patterns over time.
4. **Anomaly Detection:**
    - Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

Sample code

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose



# Load dataset

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')



# Time Series Decomposition

decomposition = seasonal_decompose(data['Page_Views'], model='additive')

decomposition.plot()
```

```
plt.show()


# Moving Average

data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()

data[['Page_Views', 'Page_Views_MA']].plot()

plt.title('Page Views with Moving Average')

plt.show()


# Anomaly Detection

data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()

anomalies = data[data['Page_Views_Z'].abs() > 3]

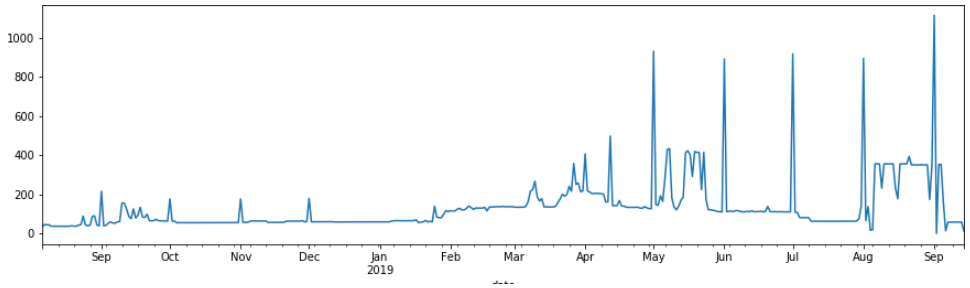
plt.plot(data.index, data['Page_Views'], label='Page Views')

plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')

plt.legend()

plt.show()
```

sample output



Expected Outcomes:

- Clear visualization of trends, seasonality, and anomalies in website traffic.
- Understanding of the underlying patterns in user behavior over time.

Experiment 7:. Random Sampling and Sampling Distribution

Objective:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Steps:

1. **Generate a Population:**
    - Create a population of data with a specified distribution (e.g., normal distribution).
2. **Random Sampling:**
    - Perform random sampling from the population to create multiple samples of different sizes.
    - Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. **Sampling Distribution:**
    - Plot histograms or density plots of sample statistics (e.g., sample means).
    - Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. **Central Limit Theorem (Optional):**
    - Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Example Code:

python

Copy code

```
import numpy as np
import matplotlib.pyplot as plt


# Step 1: Generate a population (e.g., normal distribution)
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)


# Step 2: Random sampling
sample_sizes = [30, 50, 100]  # different sample sizes to consider
num_samples = 1000  # number of samples for each sample size
```

```
sample_means = {}

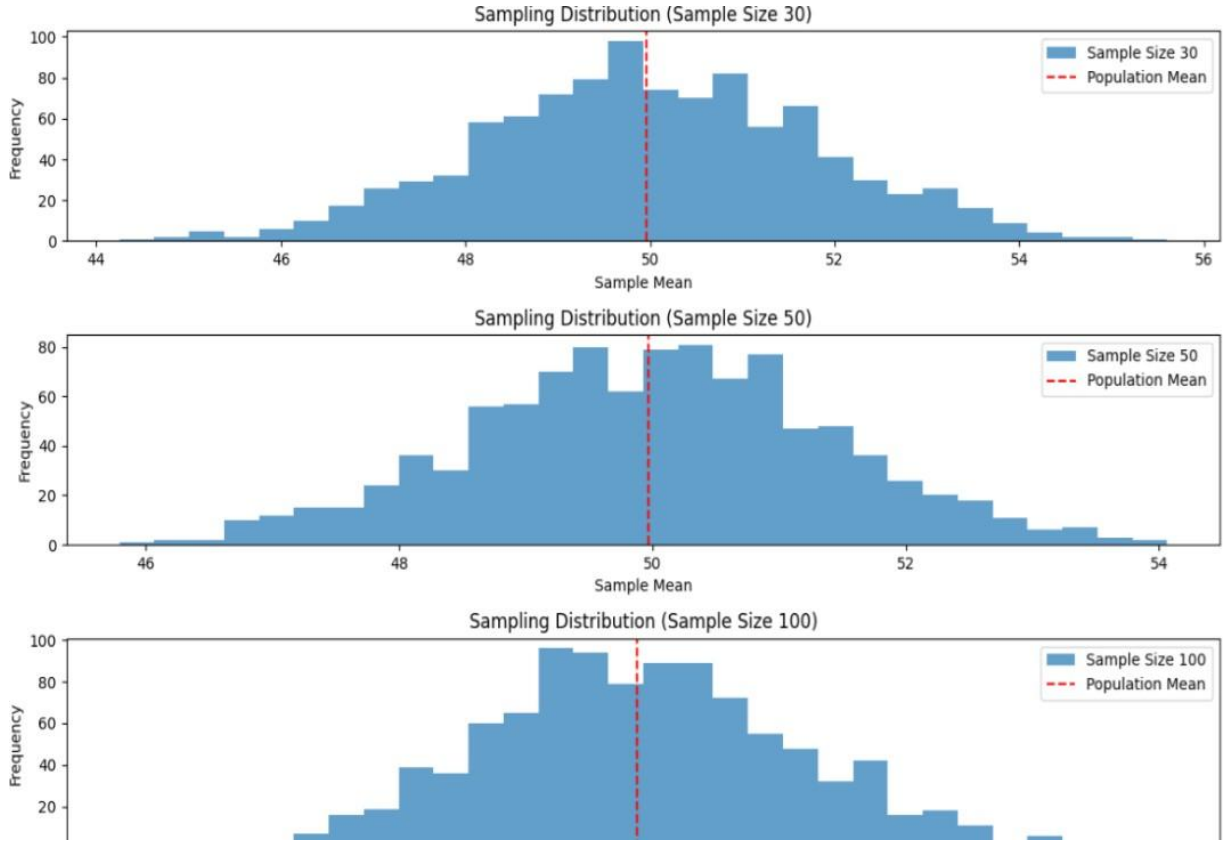
for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i+1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Population
Mean')
    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()
```

OUTPUT:


Sampling Distribution (Sample Size 30)
Frequency
Sample Mean
Sample Size 30
Population Mean
Sampling Distribution (Sample Size 50)
Frequency
Sample Mean
Sample Size 50
Population Mean
Sampling Distribution (Sample Size 100)
Frequency
Sample Size 100
Population Mean

Lab Experiment 8: Conduct Z test for the Data Given using Python Objective:

To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

1. **Null Hypothesis (H_0):** The average weight of the birds is 150 grams.
2. **Alternative Hypothesis (H_1):** The average weight of the birds is not 150 grams.
3. **Sample:** Measure the weights of 30 birds randomly selected from the population.
4. **Z-Test:** Conduct a Z-test to compare the sample mean to 150 grams.
5. **Decision Rule:** Use a significance level of $\alpha = 0.05$.

Python Code Example:

python

Copy code

```
import numpy as np

import scipy.stats as stats


# Define the sample data (hypothetical weights in grams)

sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151, 150, 149, 152, 151, 148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])

# Population mean under the null hypothesis

population_mean = 150

# Calculate sample statistics

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1)  # Using sample standard deviation


# Number of observations

n = len(sample_data)

# Calculate the Z-statistic

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))


# Calculate the p-value
```

```
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))  # Two-tailed test


# Print results

print(f"Sample Mean: {sample_mean:.2f}")

print(f"Z-Statistic: {z_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level

alpha = 0.05

if p_value < alpha:

    print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")

else:

    print("Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.")
```

OUTPUT:

```
Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.
```

Exp No:**9** Experiment to understand Matplotlib library use cases in Data Science through visualization

Description: Use Iris data set to understand the Matplot lib library

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline


data=pd.read_csv('/content/Iris_Dataset.csv')
data
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

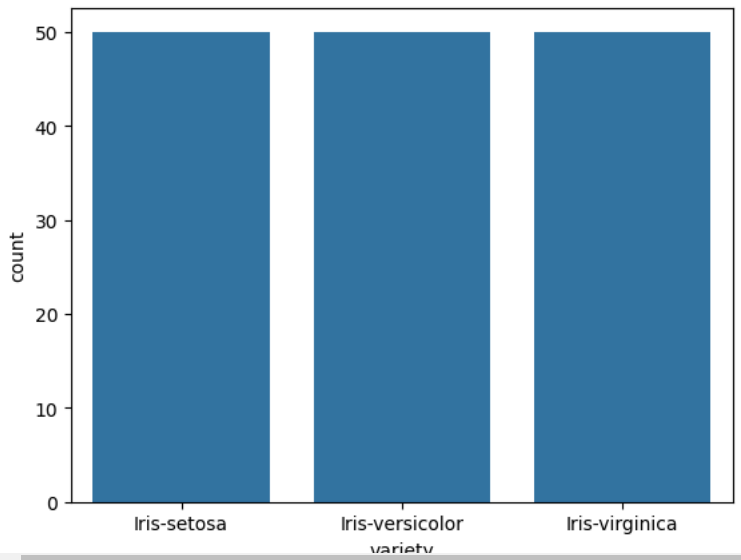
```
data.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
data.value_counts('variety')
```

variety	count
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

```
sns.countplot(x='variety',data=data,)
plt.show()
```




```
dummies=pd.get_dummies(data.variety)
```

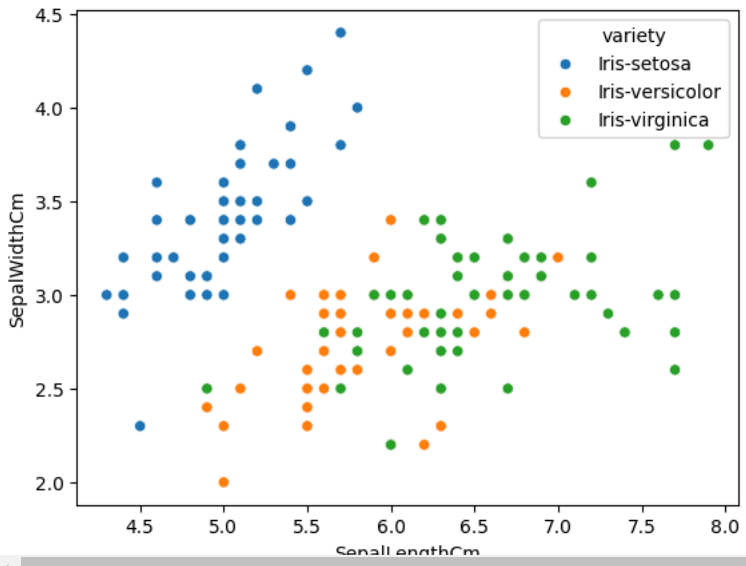
```
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)
```

```
FinalDataset.head()
```

	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5
4	True	False	False	5	5.0	3.6	1.4

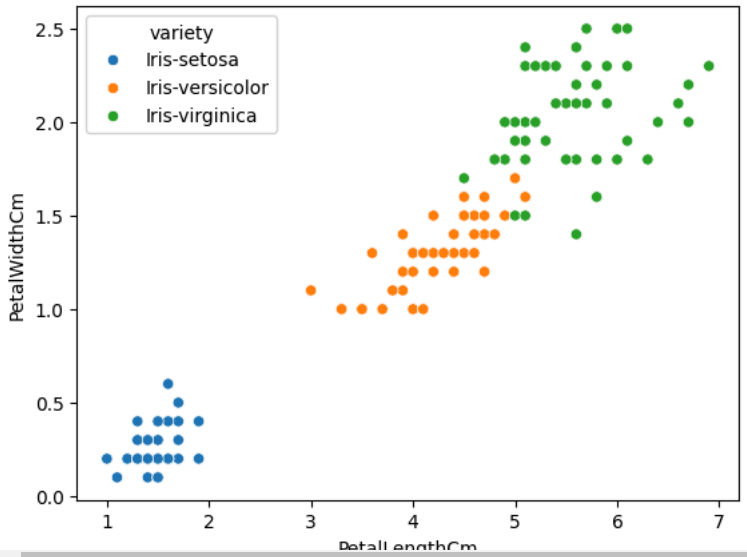
```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)
```

```
<Axes: xlabel='SepalLengthCm', ylabel='SepalWidthCm'>
```

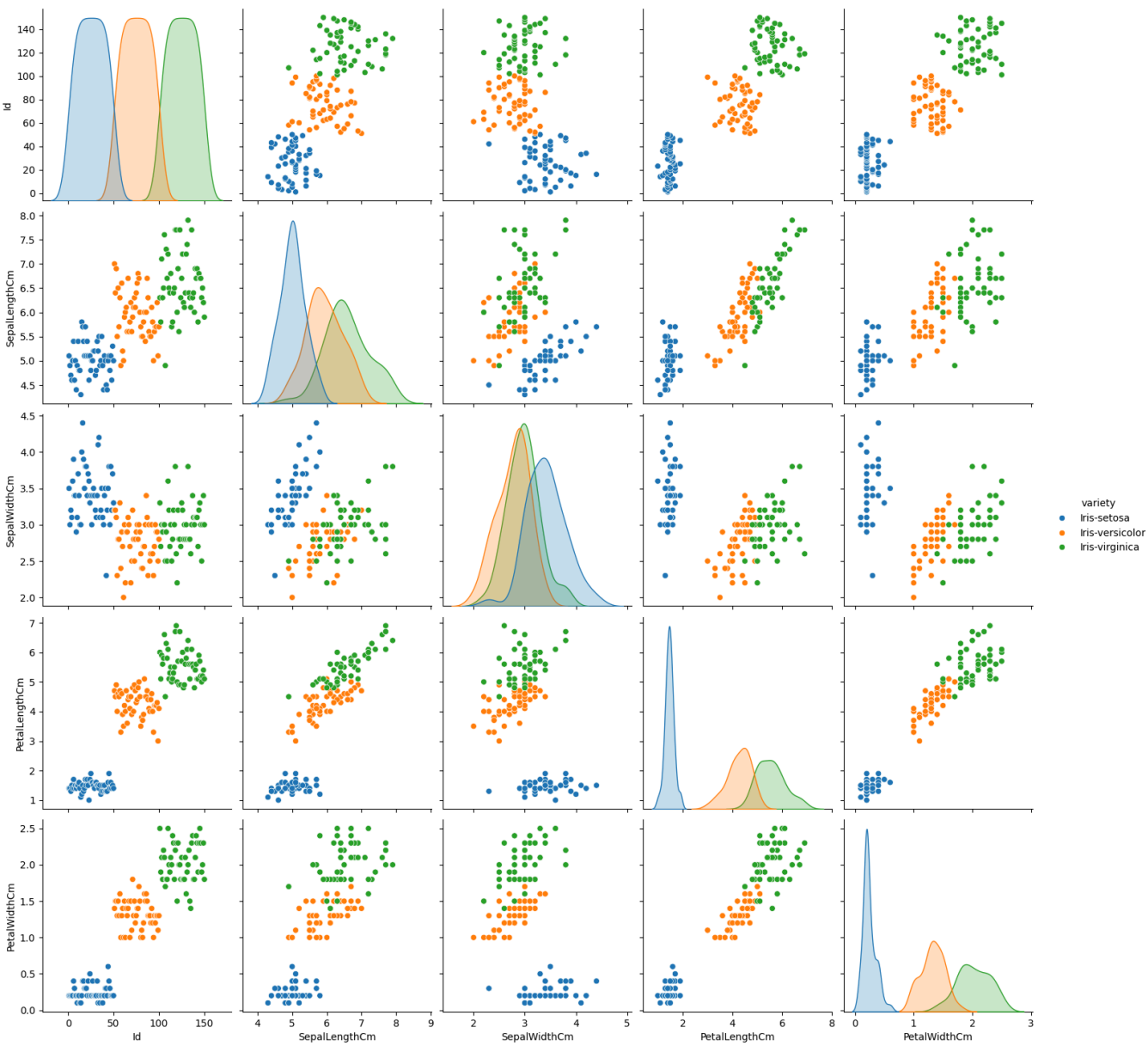



```
sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)
```

```
<Axes: xlabel='PetalLengthCm', ylabel='PetalWidthCm'>
```

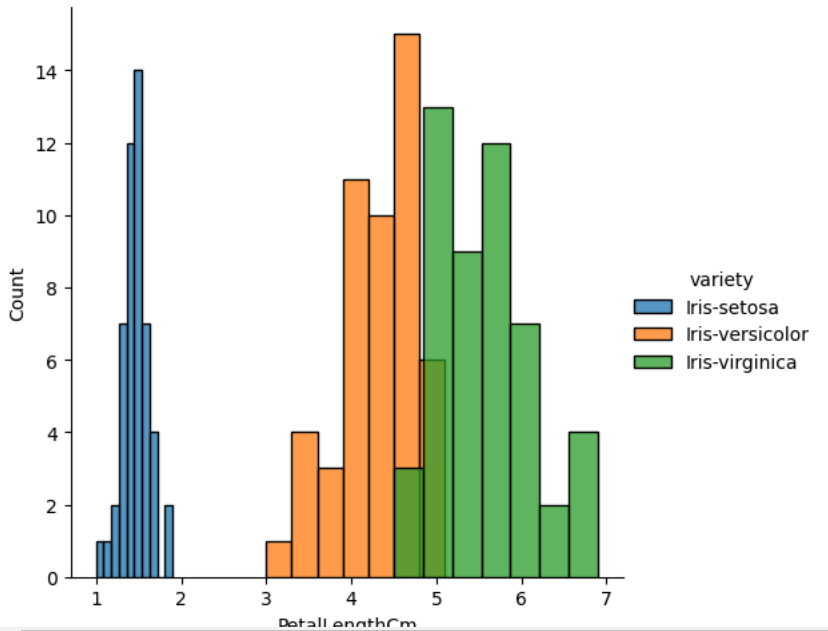



```
sns.pairplot(data,hue='variety',height=3);
```

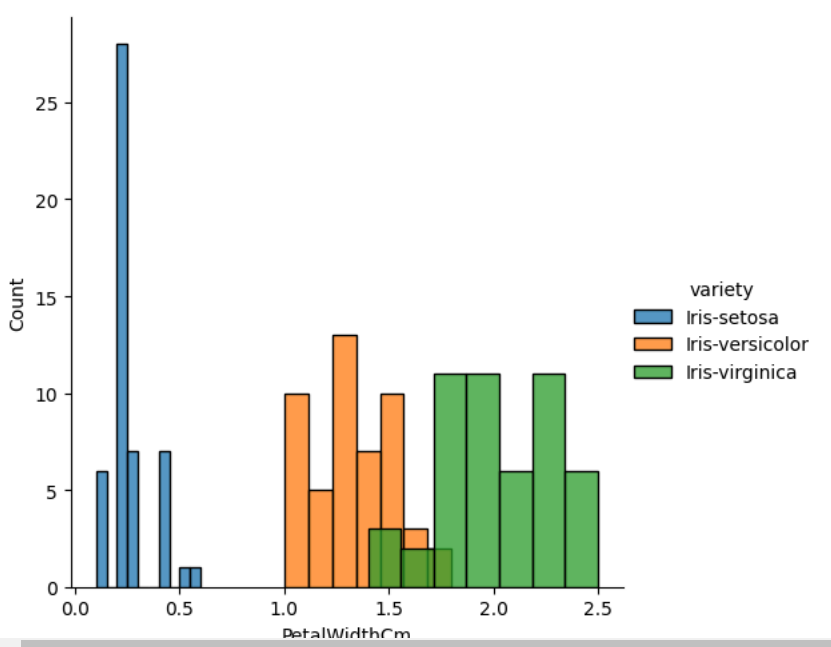


```
plt.show()
```

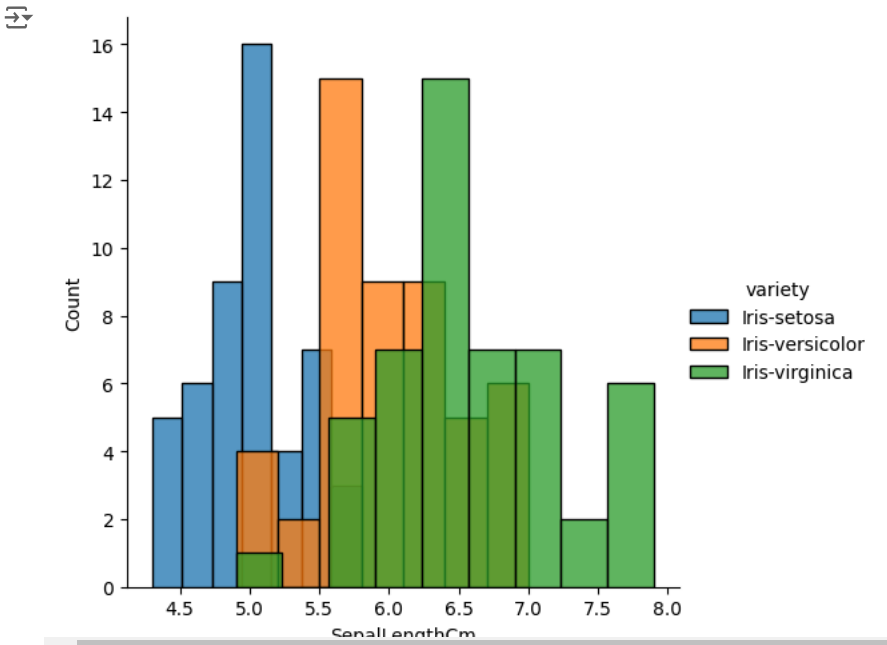
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();
plt.show();
```



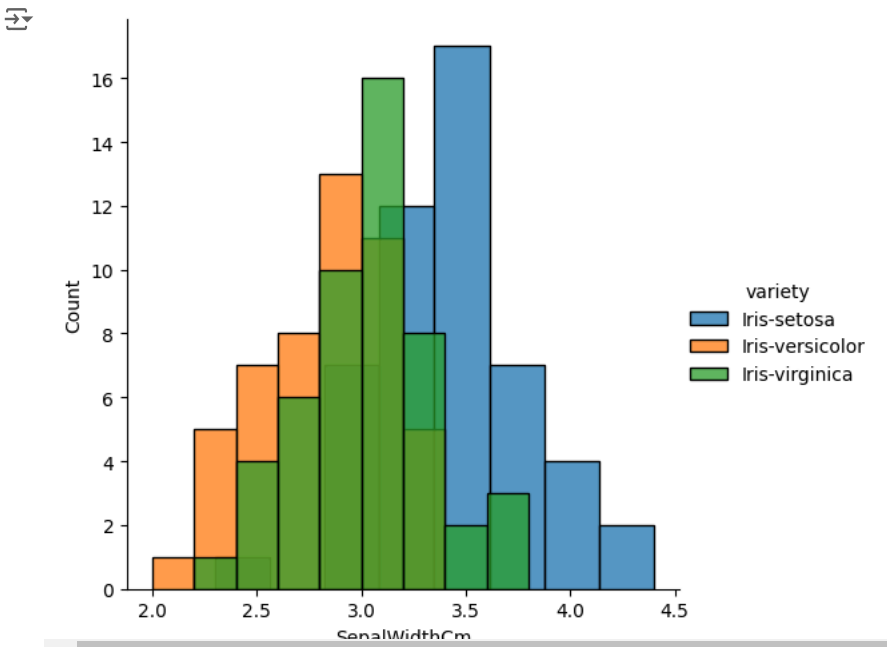

```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();
plt.show();
```




```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();
plt.show();
```




```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();
plt.show();
```



Exp No:**10** Experiment to understand array function in Data science.

Description: Understand array function using Numpy library

```
import numpy as np
array=np.random.randint(1,100,9)
array
```

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
```

```
np.sqrt(array)
```

```
array([9.11043358, 5.        , 4.35889894, 6.8556546 , 7.87400787,
       3.87298335, 9.79795897, 6.244998  , 7.14142843])
```

```
array.ndim
```

```
1
```

```
new_array=array.reshape(3,3)
```

```
new_array
```

```
array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])
```

```
new_array.ndim
```

```
2
```

```
new_array.ravel()
```

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
```

```
newm=new_array.reshape(3,3)
```

```
newm
```

```
array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])
```

```
newm[2,1:3]
```

```
array([39, 51])
```

```
newm[1:2,1:3]
```

```
array([[62, 15]])
```

```
new_array[0:3,0:0]
```

```
array([], shape=(3, 0), dtype=int64)
```

```
new_array[0:2,0:1]
```

```
array([[83],
       [47]])
```

```
new_array[0:3,0:1]
```

```
array([[83],
       [47],
       [96]])
```

```
new_array[1:3]
```

```
array([[47, 62, 15],
       [96, 39, 51]])
```

Exp No:11 Experiment to understand pandas library use cases in Data science.

Description: Understand data frame use cases using pandas library

```
import numpy as np
import pandas as pd
list=[[1,'Smith',50000],[2,'Jones',60000]]
```

```
df=pd.DataFrame(list)
df
```

	0	1	2
0	1	Smith	50000
1	2	Jones	60000

```
df.columns=['Empd','Name','Salary']
df
```

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 176.0+ bytes
```

```
df=pd.read_csv("/content/50_Startups.csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     object
 4   Profit           50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

```
df.head()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

```
df.tail()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	45173.06	California	14681.40

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/employee.csv")
```

```
df.head()
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

```
df.tail()
```

	emp id	name	salary
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      object
 2   salary  7 non-null      int64
dtypes: int64(2), object(1)
memory usage: 296.0+ bytes
```

```
df.salary
```

	salary
0	5000
1	6000
2	7000
3	5000
4	8000
5	3000
6	6000

```
type(df.salary)
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None

One-dimensional ndarray with axis labels (including time series).

Labels need not be unique but must be a hashable type. The object
supports both integer- and label-based indexing and provides a host of
methods for performing operations involving the index. Statistical
```

```
df.salary.mean()
```

```
5714.285714285715
```

```
df.salary.median()
```

```
6000.0
```

```
df.salary.mode()
```

	salary
0	5000
1	6000

```
df.salary.var()
```

```
2571428.5714285714
```

```
df.salary.std()
```

```
1603.5674514745463
```

```
df.describe()
```

	emp id	salary
count	7.000000	7.000000
mean	4.000000	5714.285714
std	2.160247	1603.567451
min	1.000000	3000.000000
25%	2.500000	5000.000000
50%	4.000000	6000.000000
75%	5.500000	6500.000000
max	7.000000	8000.000000

```
df.describe(include='all')
```

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714
std	2.160247	NaN	1603.567451
min	1.000000	NaN	3000.000000
25%	2.500000	NaN	5000.000000
50%	4.000000	NaN	6000.000000
75%	5.500000	NaN	6500.000000
max	7.000000	NaN	8000.000000

```
empCol=df.columns
```

```
empCol
```

```
Index(['emp id', 'name ', 'salary'], dtype='object')
```

```
emparray=df.values
```

```
emparray
```

```
array([[1, 'SREE VARSSINI K S', 5000],
       [2, 'SREEMATHI B', 6000],
       [3, 'SREYA G', 7000],
       [4, 'SREYASKARI MULLAPUDI', 5000],
       [5, 'SRI AKASH U G', 8000],
```

```
            [6, 'SRI HARSHAVARDHANAN R', 3000],
            [7, 'SRI HARSHAVARDHANAN R', 6000]], dtype=object)
```

```
employee_DF=pd.DataFrame(emparray,columns=empCol)
```

```
employee_DF
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

Start coding or generate with AI.

Exp No:12 Experiment to detect outliers in a given data set.

Description: Understand the procedure to identify the outliers in a given dataset

```
#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16)  # randomly generate 16 numbers between  1 to 100
array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
array.mean()
```

```
50.5
```

```
np.percentile(array,25)
```

```
26.0
```

```
np.percentile(array,50)
```

```
56.0
```

```
np.percentile(array,75)
```

```
69.0
```

```
np.percentile(array,100)
```

```
90.0
```

```
#outliers detection
def outDetection(array):
  sorted(array)
  Q1,Q3=np.percentile(array,[25,75])
  IQR=Q3-Q1
  lr=Q1-(1.5*IQR)
  ur=Q3+(1.5*IQR)
  return lr,ur
```

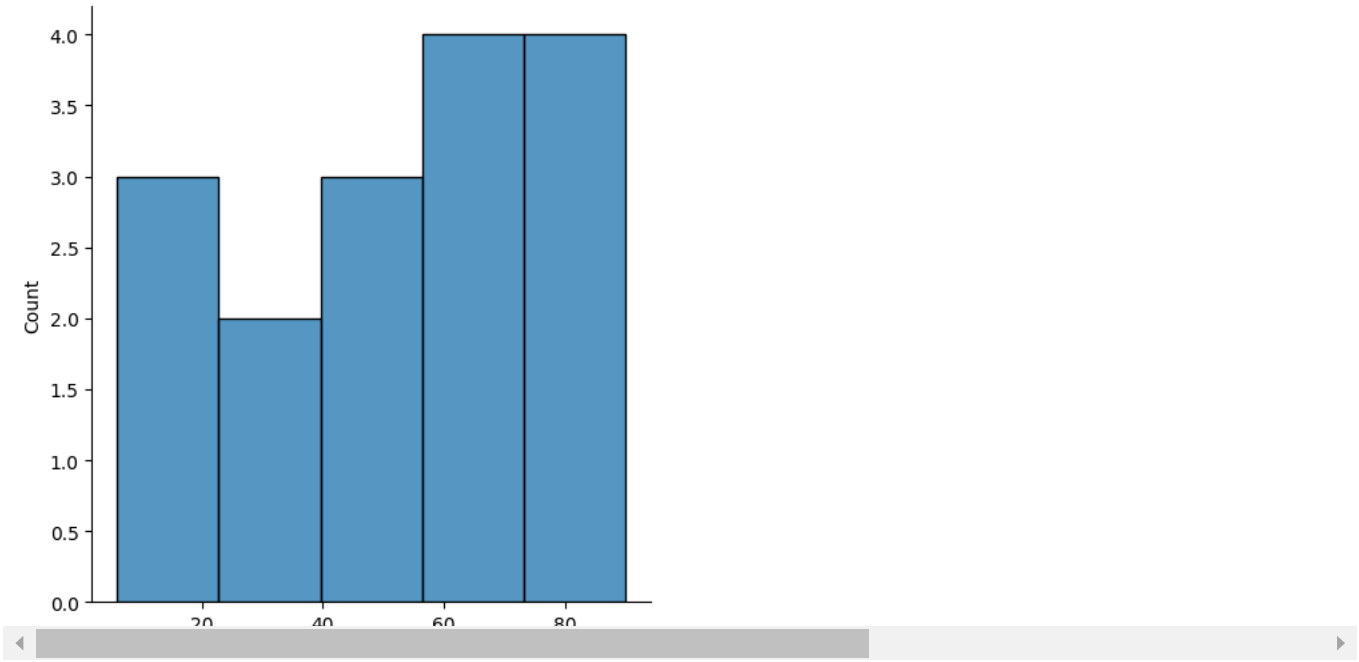
```
lr,ur=outDetection(array)
```

```
lr,ur
```

```
(-38.5, 133.5)
```

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f3291c2710>
```




```
sns.distplot(array)
```

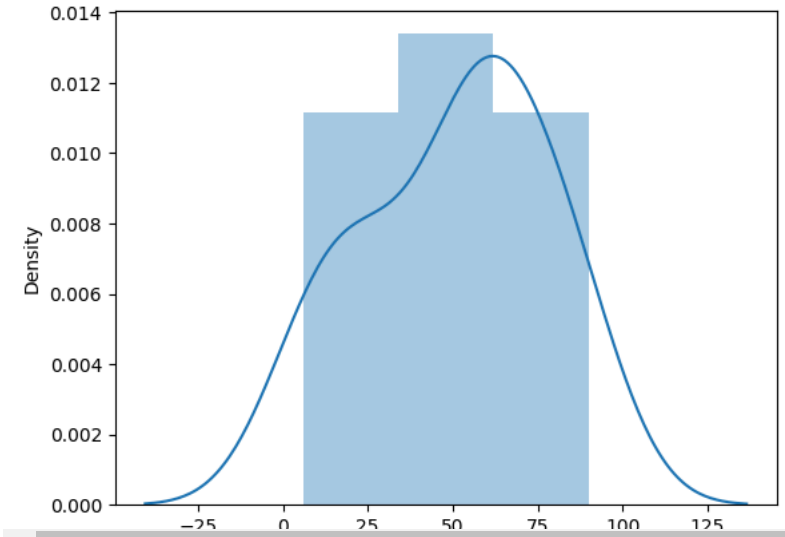
```
<ipython-input-19-d72101983c40>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(array)
<Axes: ylabel='Density'>
```

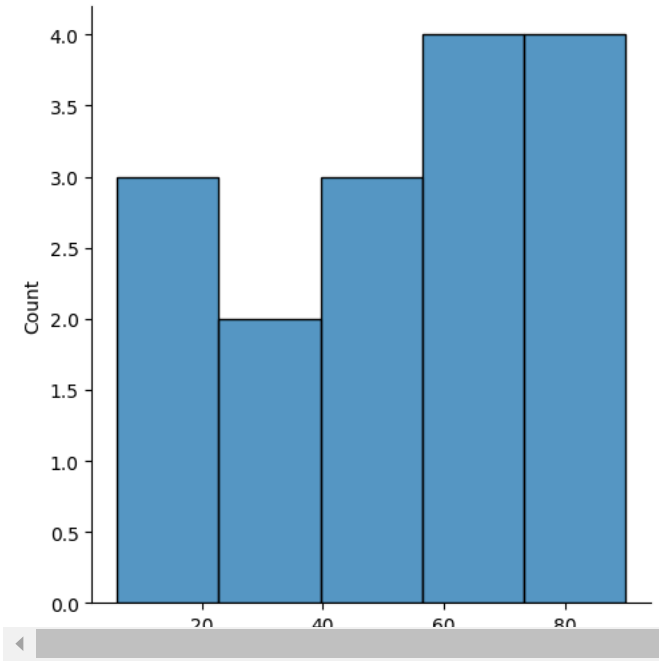



```
new_array=array[(array>lr) & (array<ur)]
new_array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
sns.displot(new_array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f2e09bb580>
```




```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

```
(-38.5, 133.5)
```

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
sns.distplot(final_array)
```

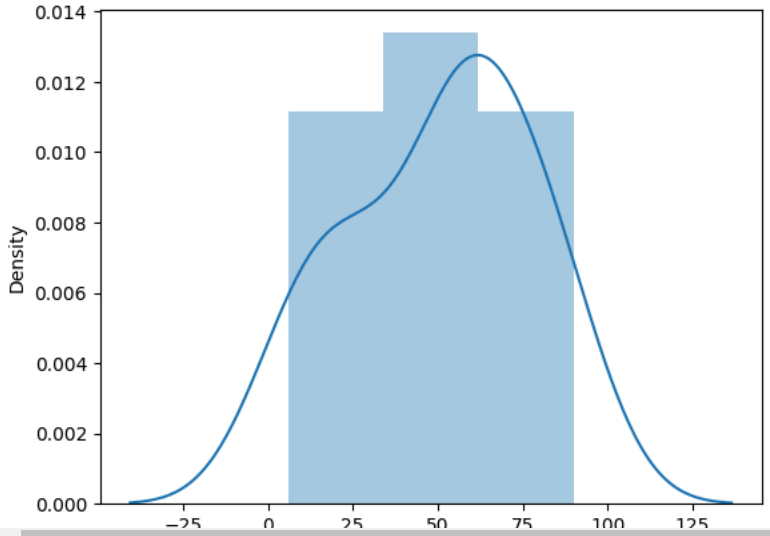
```
<ipython-input-18-7ba96ada5b76>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(final_array)
<Axes: ylabel='Density'>
```



Exp No:13 Experiment to handle missing values and inappropriate data in a given data set

.Description: Understand the procedure to handle missing and inappropriate data

Exp:6 Handling Missing and Inappropriate Data in a Dataset

Aim: Demonstrate an experiment to handle missing data and inappropriate data in a Data set using Python Pandas Library for Data Preprocessing.

Dataset Given:

Hotel.csv

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group
1	20-25	4	Ibis	veg	1300	2	40000	20-25
2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
3	25-30	6	RedFox	Veg	1322	2	30000	25-30
4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
5	35+	3	Ibis	Vegetarian	989	2	45000	35+
6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

About Dataset:

No.of Columns =9 (called as series – CustomerID, Age_Group, Rating(1-5),Hotel, FoodPreference, Bill, NoOfPax, EstimatedSalary)

CutomerID: Numerical Continuous data

Age: Categorical Data

Rating (1-5): Numerical Discrete Data

Hotel: Categorical Data

Food: Categorical Data

Bill: Numerical Continuous data

NoOfPax: Numerical Discrete

EstimatedSalary: Numerical Continuous data

Python Code:

Upload Hotel.csv and convert it into dataFrame

import numpy as np

import pandas as pd

df=pd.read_csv("Hotel_Dataset.csv")

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

#From the dataframe identify the duplicate row(i.e row 9)

The duplicated() method returns a Series with True and False values that describe which rows in the DataFrame are duplicated and not.

df.duplicated()

```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool
```

The info() method prints information about the DataFrame. The information contains the number of columns, column labels, column data types, memory usage, range index, and the number of cells in each column (non-null values).

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   CustomerID        11 non-null     int64
 1   Age_Group         11 non-null     object
 2   Rating(1-5)       11 non-null     int64
 3   Hotel             11 non-null     object
 4   FoodPreference    11 non-null     object
 5   Bill              11 non-null     int64
 6   NoOfPax           11 non-null     int64
 7   EstimatedSalary   11 non-null     int64
 8   Age_Group.1       11 non-null     object
dtypes: int64(5), object(4)
memory usage: 924.0+ bytes
```

The drop_duplicates() method removes duplicate rows.

df.drop_duplicates(inplace=True)

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estimated Salary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

#While removing duplicate record row index also removed

The len() function to return the length of an object. With a dataframe, the function returns the number of rows.

len(df)

10

#Reset the index

index=np.array(list(range(0,len(df))))

df.set_index(index,inplace=True)

index

```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

Axis refers to the dimensions of a DataFrame (index and columns) or Series (index only) Use axis=0 to apply functions row-wise along the index. Use axis=1 to apply functions column-wise across columns.

df.drop(['Age_Group.1'],axis=1,inplace=True)

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000
3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000
5	6	35+	3	Ibys	Non-Veg	1909	2	122220
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122
7	8	20-25	7	LemonTree	Veg	2999	-10	345673
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777

The function . loc is typically used for label indexing and can access multiple columns.

df.CustomerID.loc[df.CustomerID<0]=np.nan

df.Bill.loc[df.Bill<0]=np.nan

df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan

df

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:1: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame
See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df.CustomerID.loc[df.CustomerID<0]=np.nan
df.Bill.loc[df.Bill<0]=np.nan
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:2: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	veg	1300.0	2	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3	59000.0
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2	30000.0
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2	120000.0
4	5.0	35+	3.0	Ibis	Vegetarian	989.0	2	45000.0
5	6.0	35+	3.0	Ibys	Non-Veg	1909.0	2	122220.0
6	7.0	35+	4.0	RedFox	Vegetarian	1000.0	-1	21122.0
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	-10	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3	NaN
9	10.0	30-35	5.0	RedFox	non-Veg	NaN	4	87777.0

df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan

df

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2129877948.py:1: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Vegetarian	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibys	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Vegetarian	1000.0	NaN	21122.0
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	NaN	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	NaN
9	10.0	30-35	5.0	RedFox	non-Veg	NaN	4.0	87777.0

df.Age_Group.unique()

```
array(['20-25', '30-35', '25-30', '35+'], dtype=object)
```

df.Hotel.unique()

```
array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)
```

Using the inplace=True keyword in a pandas method changes the default behaviour such that the operation on the dataframe doesn't return anything, it instead 'modifies the underlying data

df.Hotel.replace(['Ibys'],'Ibis',inplace=True)

df.FoodPreference.unique

```
<bound method Series.unique of 0           veg
1       Non-Veg
2           Veg
3           Veg
4    Vegetarian
5       Non-Veg
6    Vegetarian
7           Veg
8       Non-Veg
9       non-Veg
Name: FoodPreference, dtype: object>
```

df.FoodPreference.replace(['Vegetarian','veg'],'Veg',inplace=True)

df.FoodPreference.replace(['non-Veg'],'Non-Veg',inplace=True)

Fillna is a Pandas function to fill the NA/NaN values with the specified method.

If column or feature is numerical continuous data then replace the missing(NaN) value by taking mean value.

If column or feature is numerical discrete data then replace the missing(NaN) value by taking median value.

If column or feature is non-numerical i.e Categorical data then replace the missing(NaN) value by taking mode value.

df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()),inplace=True)

df.NoOfPax.fillna(round(df.NoOfPax.median()),inplace=True)

df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)

df.Bill.fillna(round(df.Bill.mean()),inplace=True)

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	Veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	4.0	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	4.0	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-25	4.0	LemonTree	Veg	2999.0	2.0	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5.0	RedFox	Non-Veg	1801.0	4.0	87777.0

Exp No:14 Experiment to understand feature scaling.

Description: Understand the importance of feature scaling

```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/pre-process_datasample.csv')
```

```
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.head()
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
features=df.iloc[:,:-1].values
```

```
<ipython-input-5-20665a0bbaa1>:1: FutureWarning: A value is trying to be set on a copy of a DataFrame c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate ob

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inpla


  df.Country.fillna(df.Country.mode()[0],inplace=True)
```

```
label=df.iloc[:,-1].values
```

```
Start coding or generate with AI.
```

```
from sklearn.impute import SimpleImputer
```

```
age=SimpleImputer(strategy="mean",missing_values=np.nan)
```

```
Salary=SimpleImputer(strategy="mean",missing_values=np.nan)
```

```
age.fit(features[:,[1]])
```

```
SimpleImputer()
```

```
Salary.fit(features[:,[2]])
```

```
SimpleImputer()
```

```
SimpleImputer()
```

```
features[:,[1]]=age.transform(features[:,[1]])
```

```
features[:,[2]]=Salary.transform(features[:,[2]])
```

```
features
```

```
array([['France', 44.0, 72000.0],
       ['Spain', 27.0, 48000.0],
       ['Germany', 30.0, 54000.0],
       ['Spain', 38.0, 61000.0],
       ['Germany', 40.0, 63777.77777777778],
       ['France', 35.0, 58000.0],
       ['Spain', 38.77777777777778, 52000.0],
       ['France', 48.0, 79000.0],
       ['France', 50.0, 83000.0],
       ['France', 37.0, 67000.0]], dtype=object)
```

```
from sklearn.preprocessing import OneHotEncoder
```

```
oh = OneHotEncoder(sparse_output=False)
```

```
Country=oh.fit_transform(features[:,[0]])
```

```
Country
```

```
array([[1., 0., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [1., 0., 0.],
       [0., 0., 1.],
       [1., 0., 0.],
```

```
            [1., 0., 0.],
            [1., 0., 0.]])
```

```
final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
```

```
final_set
```

```
array([[1.0, 0.0, 0.0, 44.0, 72000.0],
       [0.0, 0.0, 1.0, 27.0, 48000.0],
       [0.0, 1.0, 0.0, 30.0, 54000.0],
       [0.0, 0.0, 1.0, 38.0, 61000.0],
       [0.0, 1.0, 0.0, 40.0, 63777.77777777778],
       [1.0, 0.0, 0.0, 35.0, 58000.0],
       [0.0, 0.0, 1.0, 38.77777777777778, 52000.0],
       [1.0, 0.0, 0.0, 48.0, 79000.0],
       [1.0, 0.0, 0.0, 50.0, 83000.0],
       [1.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)
```

```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
```

```
feat_standard_scaler
```

```
array([[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         7.58874362e-01,  7.49473254e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
        -1.71150388e+00, -1.43817841e+00],
       [-1.00000000e+00,  2.00000000e+00, -6.54653671e-01,
        -1.27555478e+00, -8.91265492e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
        -1.13023841e-01, -2.53200424e-01],
       [-1.00000000e+00,  2.00000000e+00, -6.54653671e-01,
         1.77608893e-01,  6.63219199e-16],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
        -5.48972942e-01, -5.26656882e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
         0.00000000e+00, -1.07356980e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         1.34013983e+00,  1.38753832e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         1.63077256e+00,  1.75214693e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
        -2.58340208e-01,  2.93712492e-01]])
```

```
from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
feat_minmax_scaler
```

```
array([[1.        , 0.        , 0.        , 0.73913043, 0.68571429],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 1.        , 0.        , 0.13043478, 0.17142857],
       [0.        , 0.        , 1.        , 0.47826087, 0.37142857],
       [0.        , 1.        , 0.        , 0.56521739, 0.45079365],
       [1.        , 0.        , 0.        , 0.34782609, 0.28571429],
       [0.        , 0.        , 1.        , 0.51207729, 0.11428571],
       [1.        , 0.        , 0.        , 0.91304348, 0.88571429],
       [1.        , 0.        , 0.        , 1.        , 1.        ],
       [1.        , 0.        , 0.        , 0.43478261, 0.54285714]])
```

Start coding or generate with AI.

Exp No:15 Experiment to understand the data preprocessing in Data science

Description: Understand the importance of Data preprocessing in data science

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/pre-process_datasample.csv")
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Double-click (or enter) to edit

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   Country    9 non-null      object 
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object 
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
df.Country.mode()
```

	Country
0	France

```
df.Country.mode()[0]
```

```
type(df.Country.mode())
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None
    index is not None, the resulting Series is reindexed with the index values.
dtype : str, numpy.dtype, or ExtensionDtype, optional
    Data type for the output Series. If not specified, this will be
    inferred from `data`.
    See the :ref:`user guide <basics.dtypes>` for more usages.
name : Hashable, default None
    The name to give to the Series.
```

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
```

```
df.Age.fillna(df.Age.median(),inplace=True)
```

```
df.Salary.fillna(round(df.Salary.mean()),inplace=True)
```

```
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	France	50.0	83000.0	No
9	France	37.0	67000.0	Yes

```
pd.get_dummies(df.Country)
```

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	True	False	False
9	True	False	False

```
updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)
```

```
updated_dataset
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	True	False	False	50.0	83000.0	No
9	True	False	False	37.0	67000.0	Yes

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   Country    10 non-null     object 
 1   Age        10 non-null     float64
 2   Salary     10 non-null     float64
 3   Purchased  10 non-null     object 
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
updated_dataset.Purchased.replace(['No','Yes'],[0,1],inplace=True)
```

```
updated_dataset
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

```
Start coding or generate with AI.
```

Exp No:**15** Experiment to understand EDA-Quantitative and Qualitative analysis.

Description: Understand the importance of EDA-Quantitative and Qualitative analysis.

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

```
tips=sns.load_dataset('tips')
```

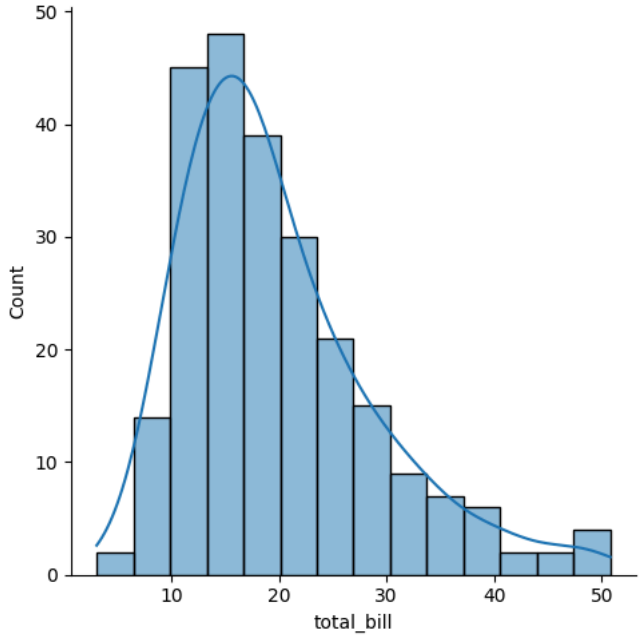
```
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

+ Code + Text

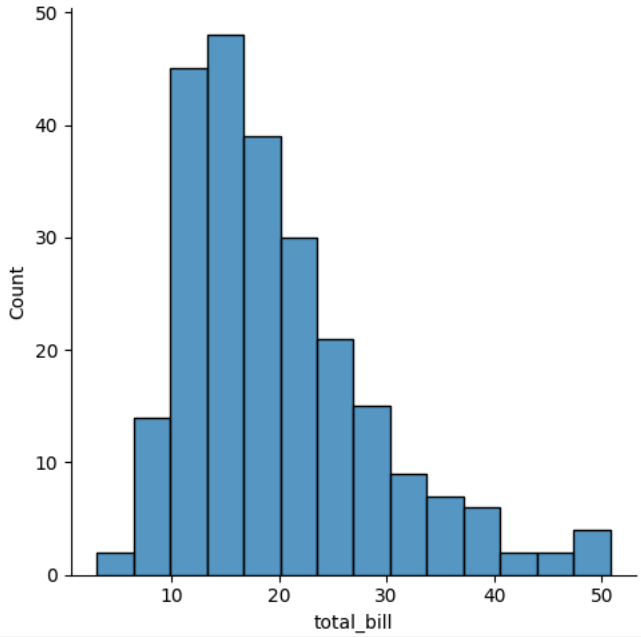
```
sns.displot(tips.total_bill,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb4c7ea680>
```



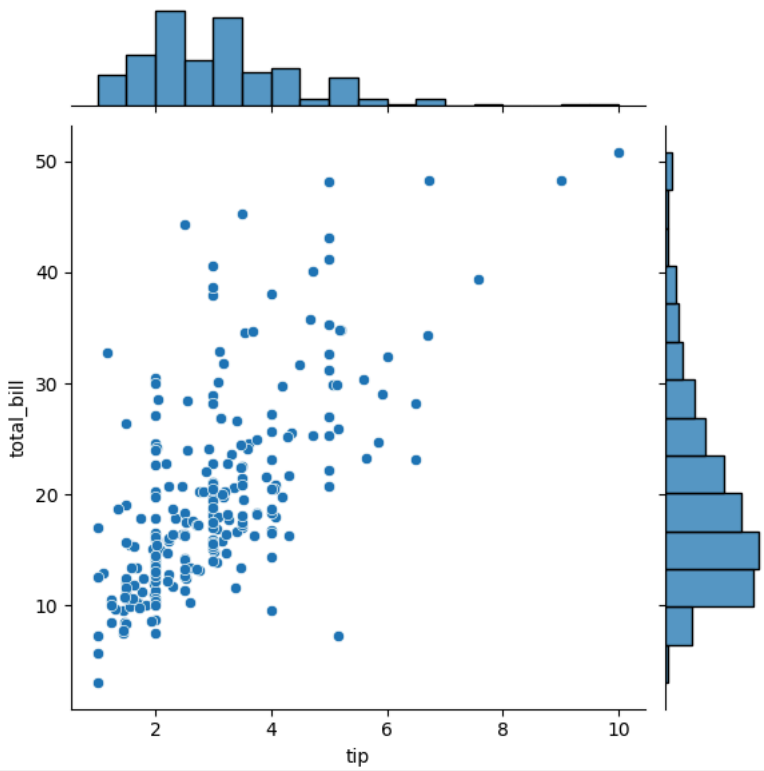

```
sns.displot(tips.total_bill,kde=False)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb0b0af580>
```



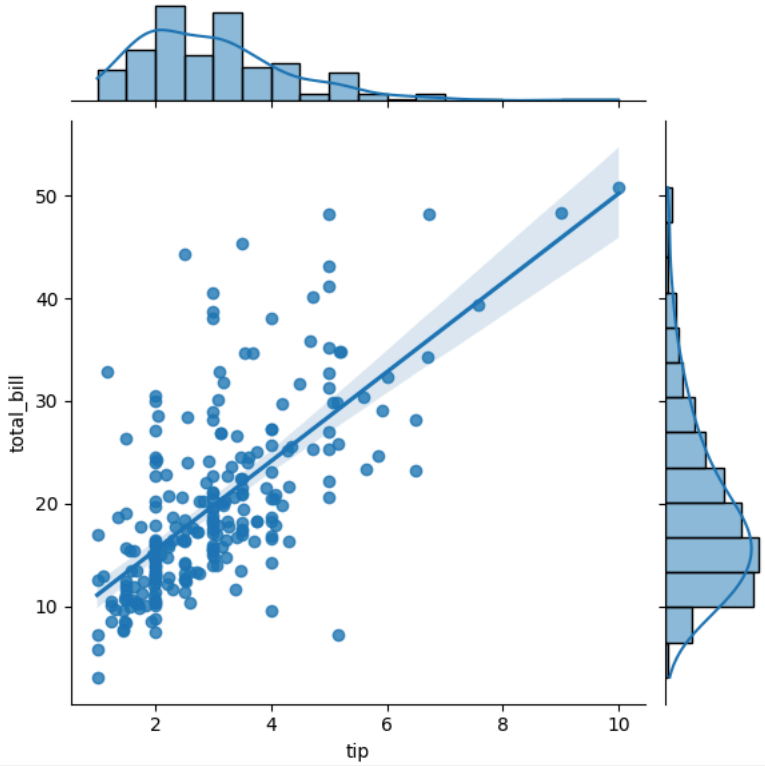

```
sns.jointplot(x=tips.tip,y=tips.total_bill)
```

```
<seaborn.axisgrid.JointGrid at 0x79bb08fc96c0>
```



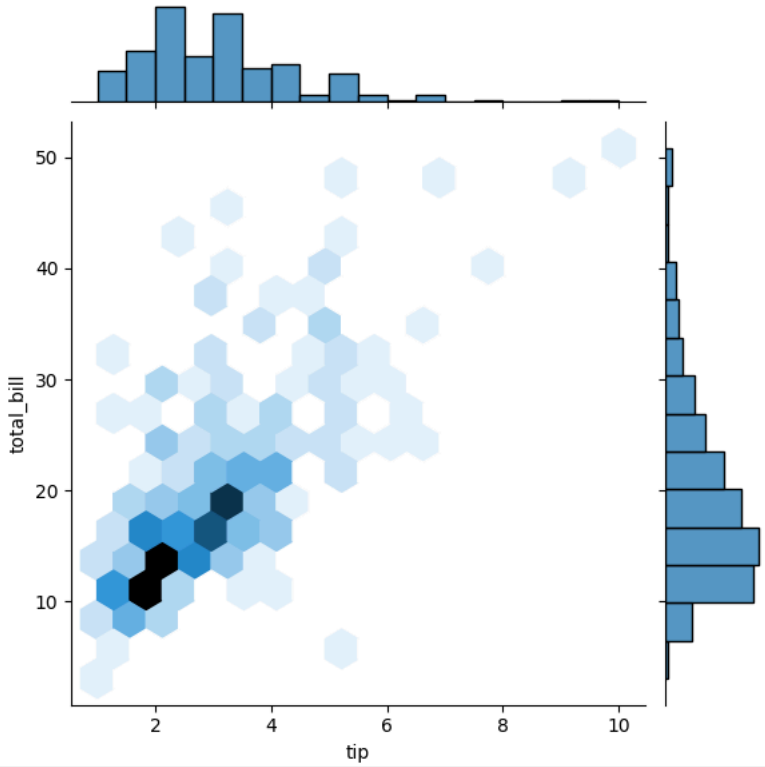

```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
```

```
<seaborn.axisgrid.JointGrid at 0x79bb08fc9cf0>
```



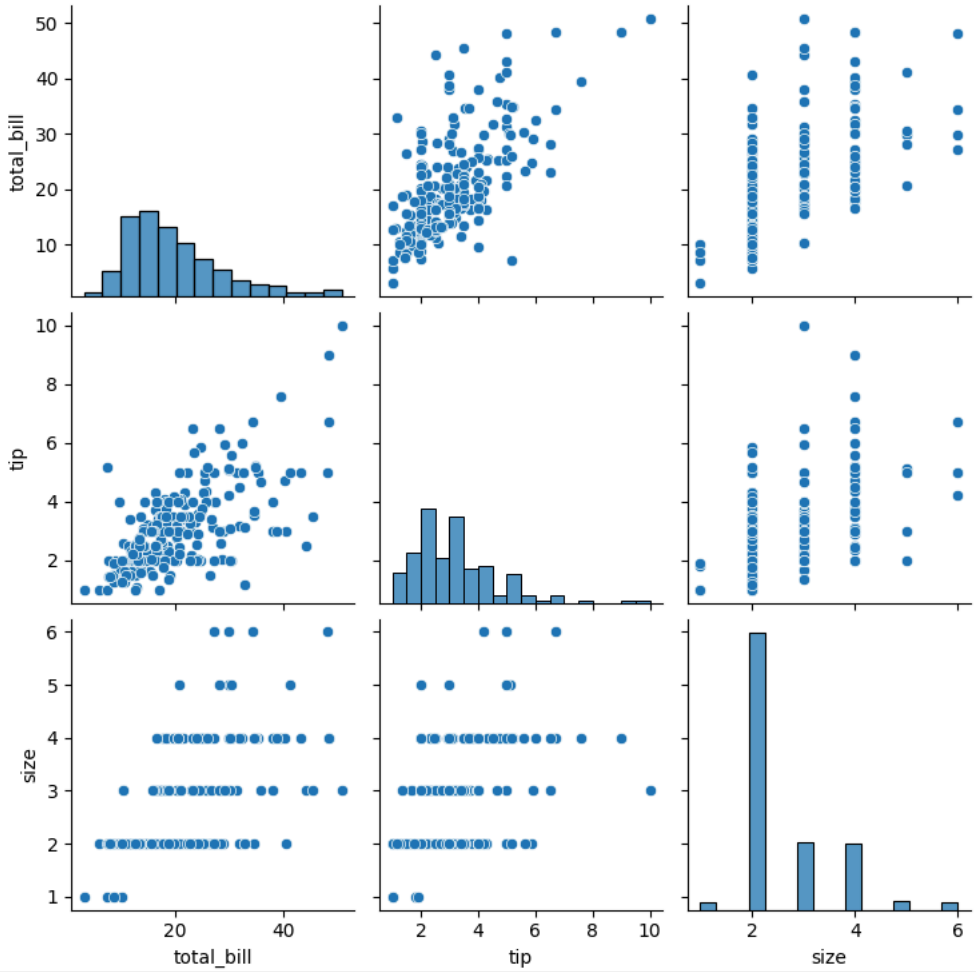

```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
```

```
<seaborn.axisgrid.JointGrid at 0x79bb088f4730>
```




```
sns.pairplot(tips)
```

```
<seaborn.axisgrid.PairGrid at 0x79bb06fc3d30>
```



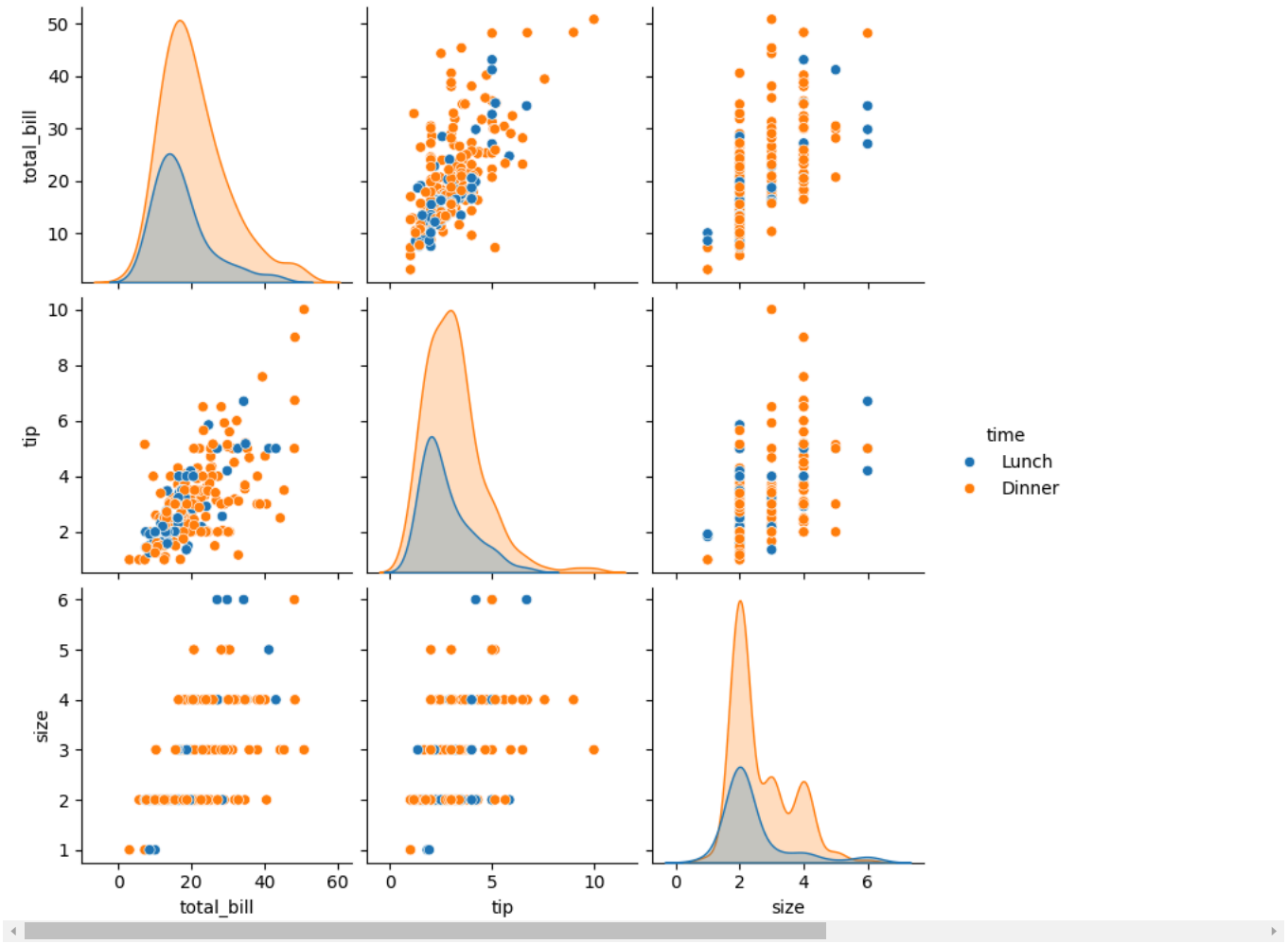

```
tips.time.value_counts()
```

time	count
Dinner	176
Lunch	68

dtype: int64

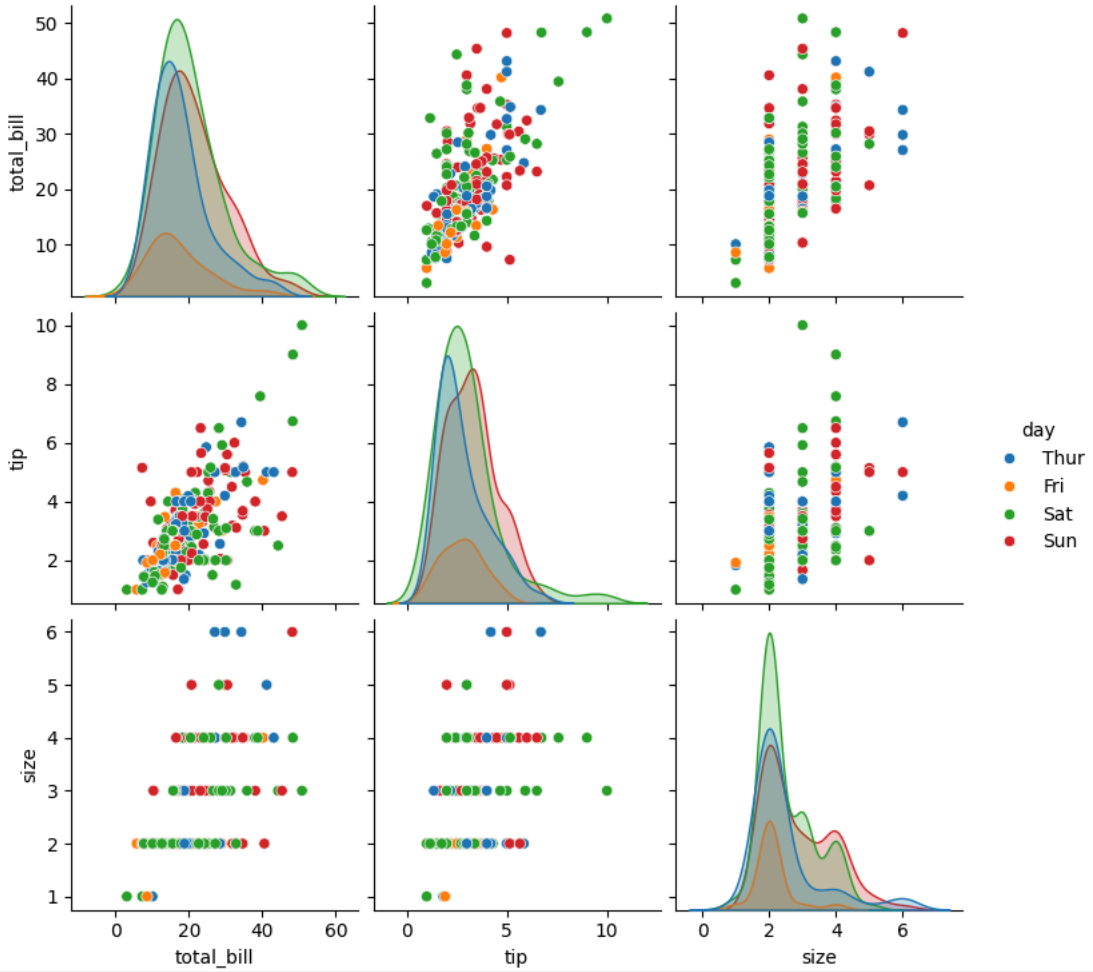
```
sns.pairplot(tips,hue='time')
```

```
<seaborn.axisgrid.PairGrid at 0x79bb088f4670>
```



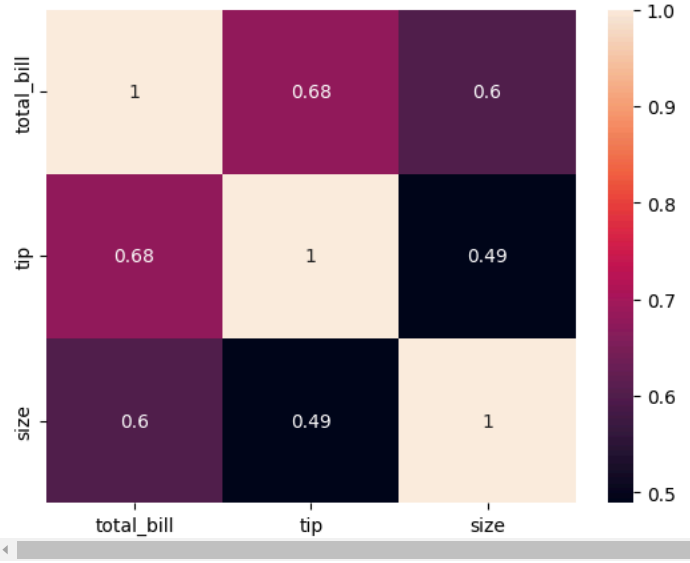

```
sns.pairplot(tips,hue='day')
```

```
<seaborn.axisgrid.PairGrid at 0x79bb08f1f6a0>
```



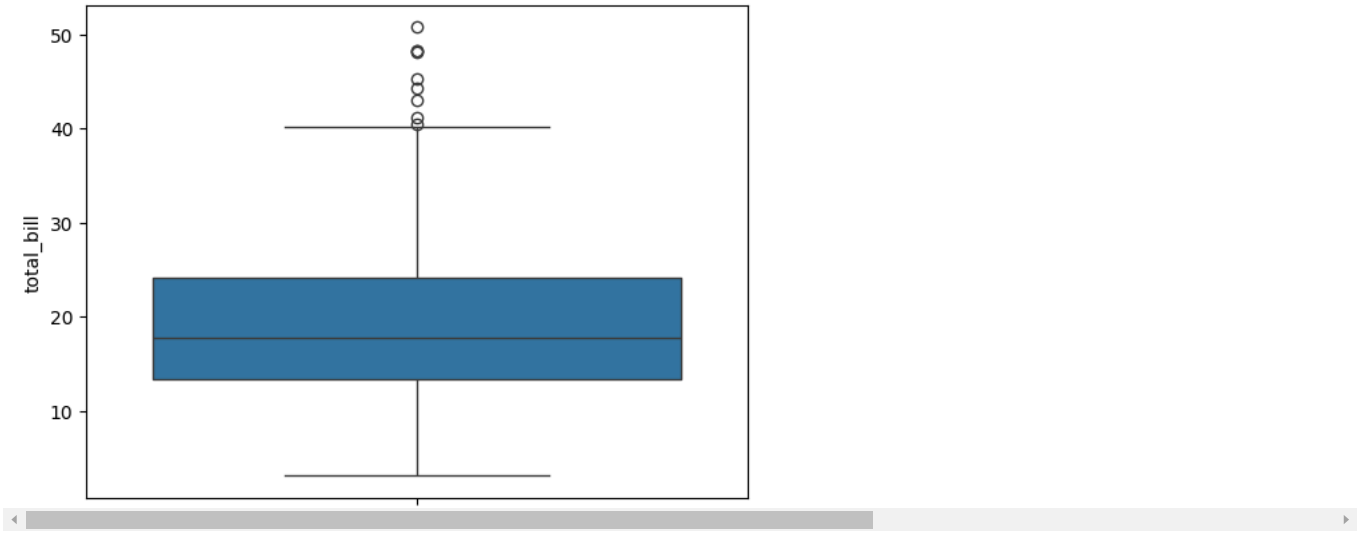

```
sns.heatmap(tips.corr(numeric_only=True),annot=True)
```

```
<Axes: >
```



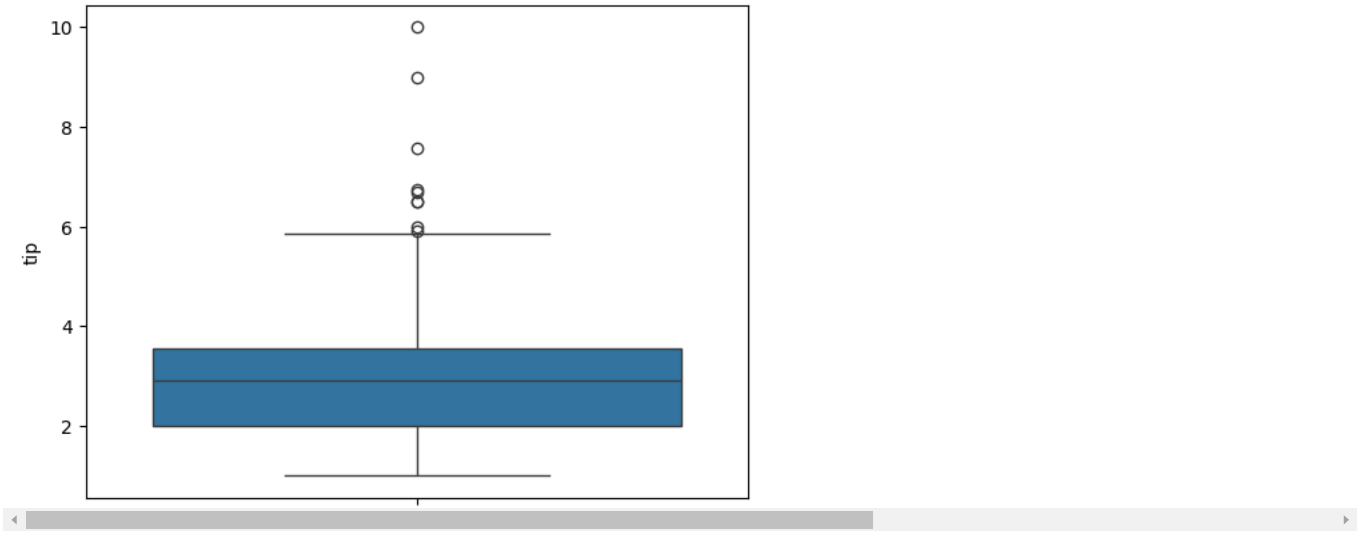

```
sns.boxplot(tips.total_bill)
```

```
<Axes: ylabel='total_bill'>
```



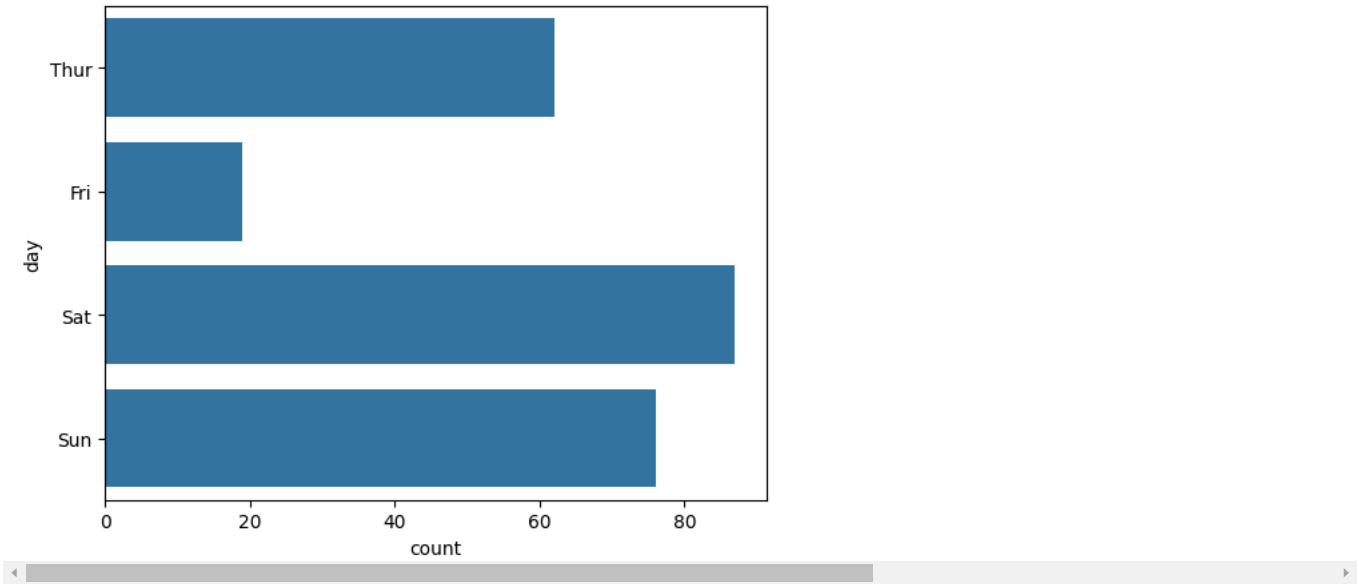

```
sns.boxplot(tips.tip)
```

```
<Axes: ylabel='tip'>
```



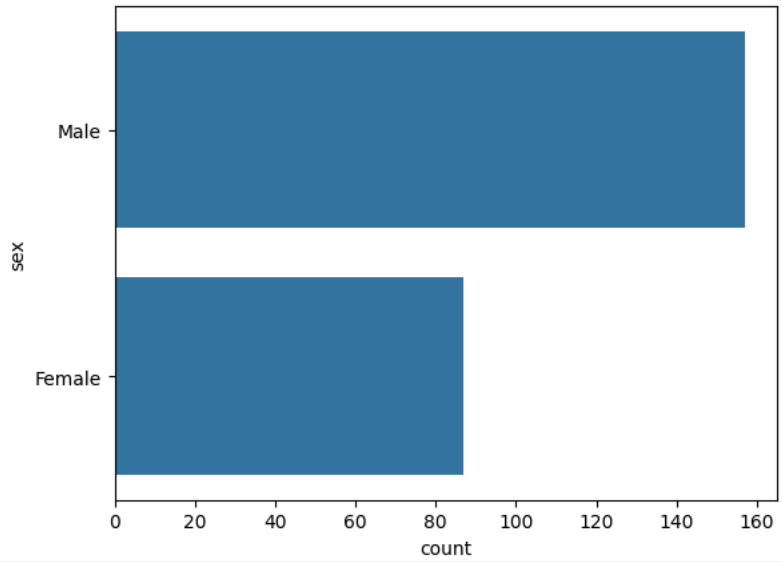

```
sns.countplot(tips.day)
```

```
<Axes: xlabel='count', ylabel='day'>
```



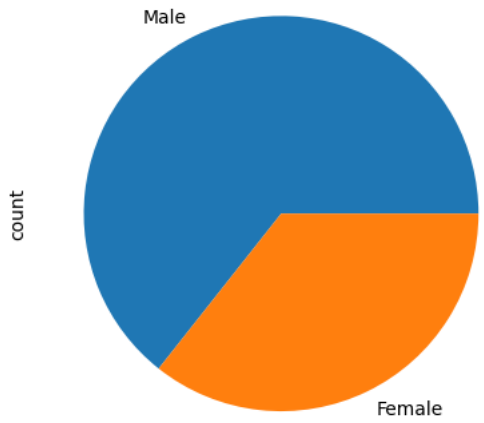

```
sns.countplot(tips.sex)
```

```
<Axes: xlabel='count', ylabel='sex'>
```



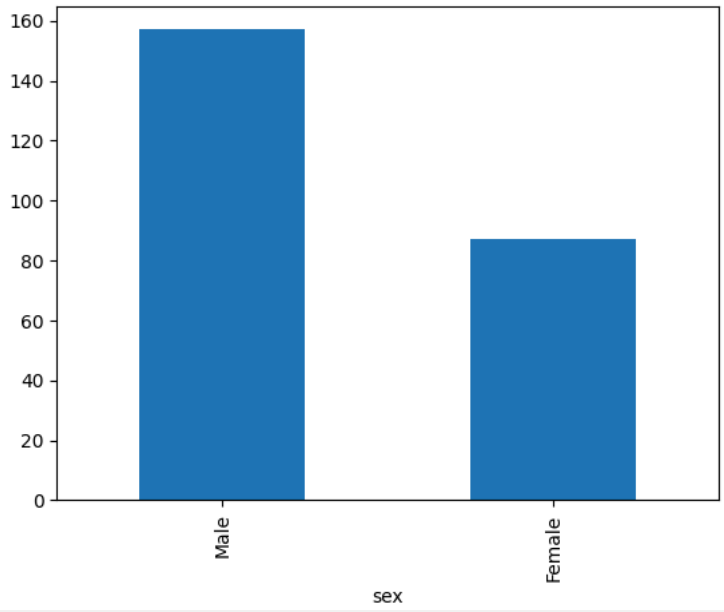

```
tips.sex.value_counts().plot(kind='pie')
```

```
<Axes: ylabel='count'>
```




```
tips.sex.value_counts().plot(kind='bar')
```

```
<Axes: xlabel='sex'>
```




```
sns.countplot(tips[tips.time=='Dinner']['day'])
```

```
<Axes: xlabel='count', ylabel='day'>
```



Exp No:**17** Experiment to understand Linear Regression for a given data set.

Description: Understand the Linear regression for the dataset given.

In [ ]:
```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

In [19]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

In [3]:
```
df.dropna(inplace=True)
```

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

In [5]:
```
df.describe()
```

Out[5]:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

In [6]:
```
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

In [7]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
```

In [20]:
```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

Out[20]:
```
LinearRegression
LinearRegression()
```

In [21]:
```
model.score(x_train,y_train)
```

Out[21]: 0.9603182547438908

In [23]:
```
model.score(x_test,y_test)
```

Out[23]: 0.9184170849214232

In [24]:
```
model.coef_
```

Out[24]: array([[9281.30847068]])

In [25]:
```
model.intercept_
```

Out[25]: array([27166.73682891])

In [26]:
```
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
```

In [27]:
```
model=pickle.load(open('SalaryPred.model','rb'))
```

In [28]:
```
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
```

```
Enter Years of Experience: 44
```

In [ ]:

In [29]:
```
print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary)
```

```
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

In [ ]:

Exp No:**18** Experiment to understand KNN algorithm for a given data set

Description: Understand the KNN algorithm for the dataset given.

In [1]:
```
import numpy as np
import pandas as pd
```

In [2]:
```
df=pd.read_csv('Iris.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [3]:
```
df.variety.value_counts()
```

Out[3]:
```
Setosa        50
Versicolor    50
Virginica     50
Name: variety, dtype: int64
```

In [4]:
```
df.head()
```

Out[4]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

In [5]:
```
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
```

In [6]:
```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
```

In [8]:
```
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,randc
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
```

Out[8]:
```
KNeighborsClassifier()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [9]:
```
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
```

```
0.9583333333333334
1.0
```

In [10]:
```
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
```

Out[10]:
```
array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  2, 48]], dtype=int64)
```

In [11]:
```
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

In [ ]:
```

```

Exp No:**19** Experiment to understand Logistic Regression for a given data set.

Description: Understand the Logistic Regression algorithm for the dataset given.

In [1]:
```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
df
```

Out[1]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [2]:
```
df.head()
```

Out[2]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

In [4]:
```
features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
features
```

Out[4]:
```
array([[    19,  19000],
       [    35,  20000],
       [    26,  43000],
       [    27,  57000],
       [    19,  76000],
       [    27,  58000],
       [    27,  84000],
       [    32, 150000],
       [    25,  33000],
       [    35,  65000],
       [    26,  80000],
       [    26,  52000],
       [    20,  86000],
       [    32,  18000],
       [    18,  82000],
       [    29,  80000],
       [    47,  25000],
       [    45,  26000],
       [    46,  28000],
```

In [5]:
```
label
```

Out[5]:
```
array([0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1,
       0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0,
       1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0,
       1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1,
       0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1,
       1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1,
       0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0,
       1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1,
       0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1,
       1, 1, 0, 1], dtype=int64)
```

In [6]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

In [7]:
```
for i in range(1,401):
  x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.
  model=LogisticRegression()
  model.fit(x_train,y_train)
  train_score=model.score(x_train,y_train)
  test_score=model.score(x_test,y_test)
  if test_score>train_score:
    print("Test {} Train{} Random State {}".format(test_score,train_score,i)
```

```
Test 0.6875 Train0.63125 Random State 3
Test 0.7375 Train0.61875 Random State 4
Test 0.6625 Train0.6375 Random State 5
Test 0.65 Train0.640625 Random State 6
Test 0.675 Train0.634375 Random State 7
Test 0.675 Train0.634375 Random State 8
Test 0.65 Train0.640625 Random State 10
Test 0.6625 Train0.6375 Random State 11
Test 0.7125 Train0.625 Random State 13
Test 0.675 Train0.634375 Random State 16
Test 0.7 Train0.628125 Random State 17
Test 0.7 Train0.628125 Random State 21
Test 0.65 Train0.640625 Random State 24
Test 0.6625 Train0.6375 Random State 25
Test 0.75 Train0.615625 Random State 26
Test 0.675 Train0.634375 Random State 27
Test 0.7 Train0.628125 Random State 28
Test 0.6875 Train0.63125 Random State 29
Test 0.6875 Train0.63125 Random State 31
```

In [8]:
```
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,
finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)
```

Out[8]:
```
LogisticRegression()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [9]:
```
print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))
```

```
0.834375
0.9125
```

In [10]:
```
from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

```
              precision    recall  f1-score   support

           0       0.85      0.93      0.89       257
           1       0.84      0.71      0.77       143

    accuracy                           0.85       400
   macro avg       0.85      0.82      0.83       400
weighted avg       0.85      0.85      0.85       400
```

In [ ]:

Exp No:**20** Experiment to understand K-means clustering algorithm for a given data set.

Description: Understand the K-means clustering algorithm for the dataset given.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

In [2]:
```
df=pd.read_csv('Mall_Customers.csv')
```

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype 
---  ------                  --------------  ----- 
 0   CustomerID              200 non-null    int64 
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64 
 3   Annual Income (k$)      200 non-null    int64 
 4   Spending Score (1-100)  200 non-null    int64 
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

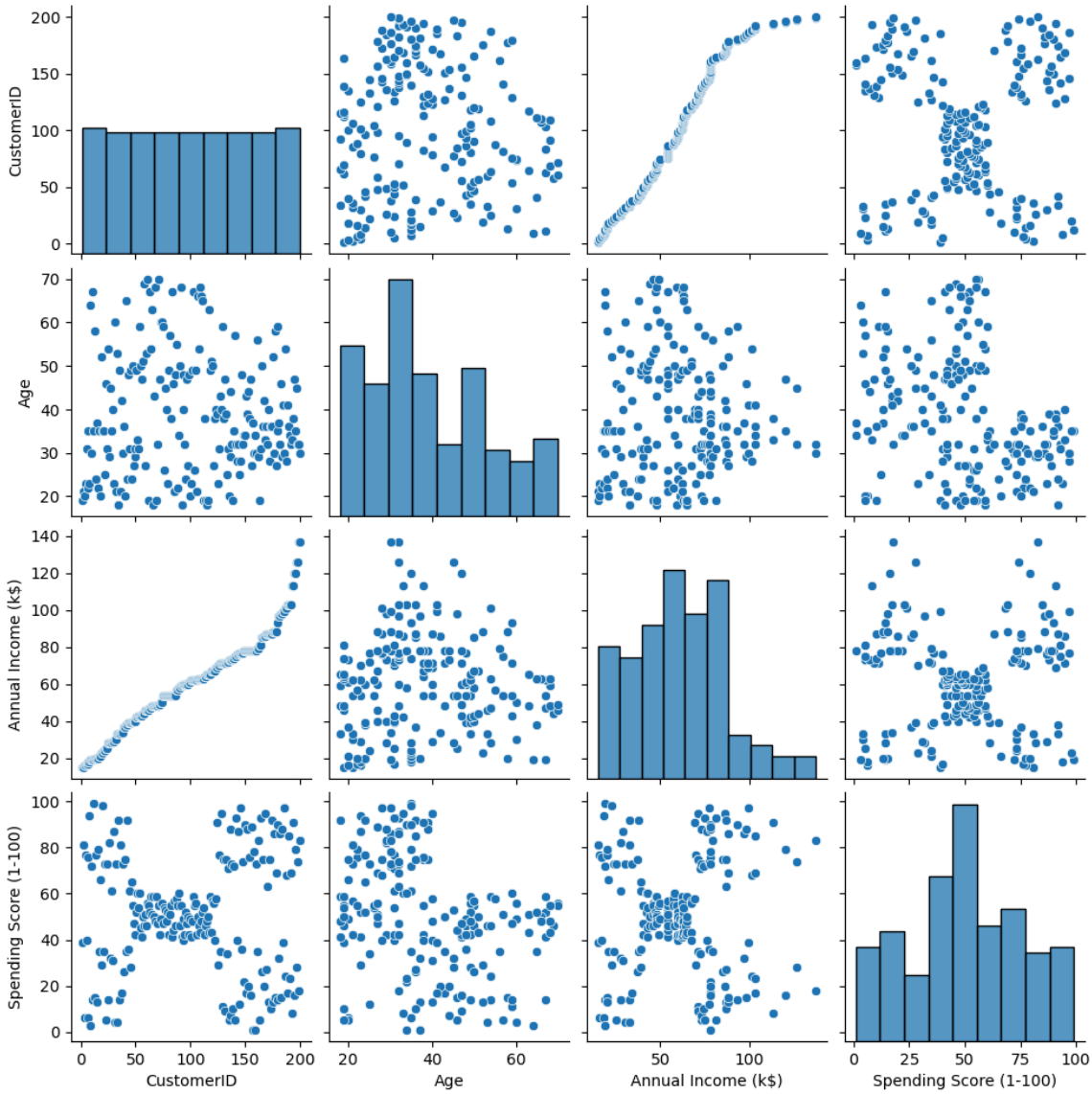
In [4]:
```
df.head()
```

Out[4]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

In [5]:
```
sns.pairplot(df)
```

Out[5]:
```
<seaborn.axisgrid.PairGrid at 0x170e8e47850>
```



In [6]:
```
features=df.iloc[:,[3,4]].values
```

In [7]:
```
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

```
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:870: FutureWarning: The default value of `n_init` will chang
e from 10 to 'auto' in 1.4. Set the value of `n_init` explicitly to suppre
ss the warning
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
Windows with MKL, when there are less chunks than available threads. You c
an avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```

Out[7]:
```
KMeans(n_clusters=5)
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [8]:
```
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
```

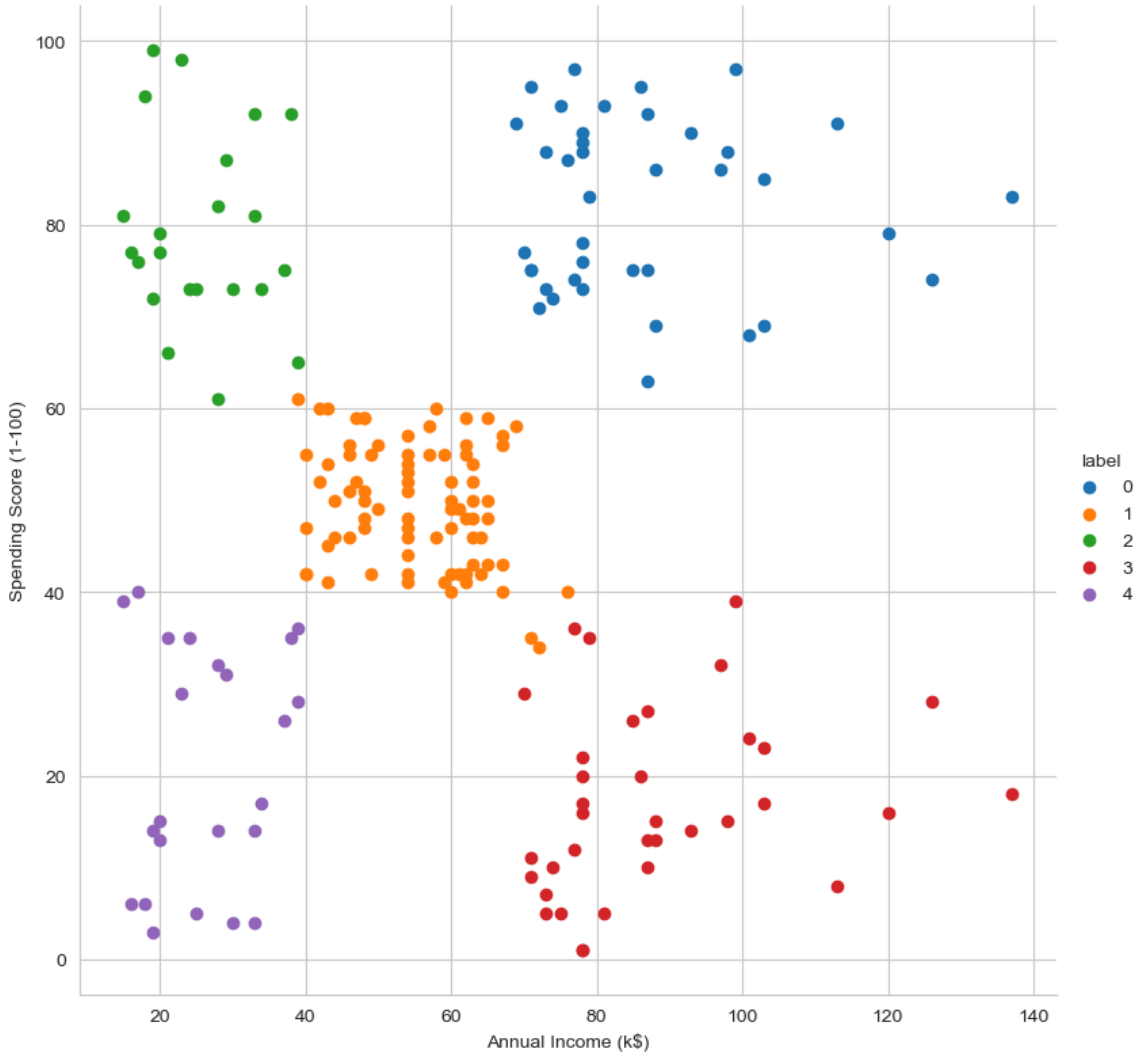
```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_8116\470183701.py:2: Setti
ngWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-doc
s/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://
pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-
view-versus-a-copy)
  Final['label']=model.predict(features)
```

Out[8]:

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

In [9]:
```
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
```

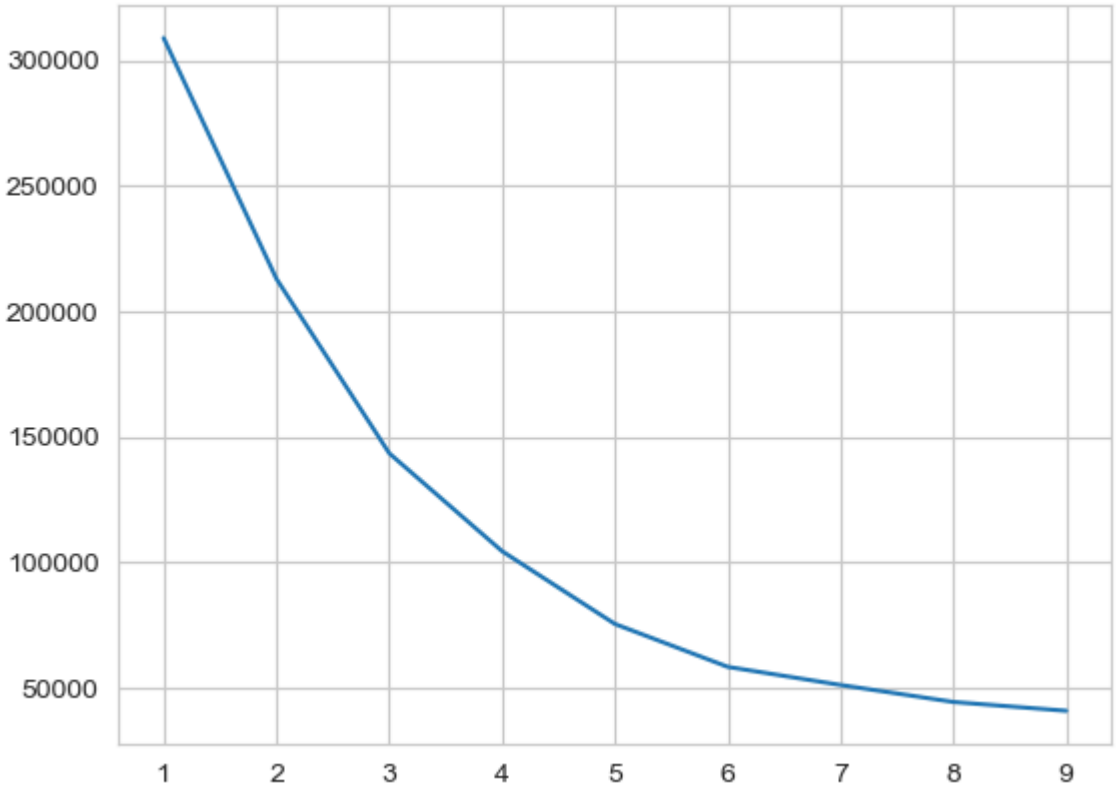


In [10]:
```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
  model=KMeans(n_clusters=i)
  model.fit(features_el)
  wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

```
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:870: FutureWarning: The default value of `n_init` will chang
e from 10 to 'auto' in 1.4. Set the value of `n_init` explicitly to suppre
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  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
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  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
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  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
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  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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ss the warning
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
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an avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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ss the warning
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
Windows with MKL, when there are less chunks than available threads. You c
an avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
```

```
er\_kmeans.py:870: FutureWarning: The default value of `n_init` will chang
e from 10 to 'auto' in 1.4. Set the value of `n_init` explicitly to suppre
ss the warning
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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Windows with MKL, when there are less chunks than available threads. You c
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  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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e from 10 to 'auto' in 1.4. Set the value of `n_init` explicitly to suppre
ss the warning
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
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ss the warning
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on
Windows with MKL, when there are less chunks than available threads. You c
an avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```

Out[10]: `[<matplotlib.lines.Line2D at 0x170e99f3550>]`



In [ ]:

VIVA - NUMPY

1. What is Numpy?

Answer: NumPy is a Python library used for working with arrays. It also provides functions for performing mathematical and logical operations on arrays.

1. How do you install NumPy?

Answer: You can install NumPy using the pip package manager with the command pip install numpy.

2. How do you import NumPy in your code?

Answer: You can import NumPy using the command import numpy as np.

3. How do you create a NumPy array from a Python list?

Answer: You can create a NumPy array using np.array([1, 2, 3]).

4. What is the difference between a list and a NumPy array?

Answer: A list is a built-in Python data structure that can hold heterogeneous items. A NumPy array is a grid of values, all of the same type, indexed by a tuple of nonnegative integers.

5. How do you create an array of zeros in NumPy?

Answer: You can create an array of zeros using np.zeros((2, 3)).

6. How do you create an array of ones in NumPy?

Answer: You can create an array of ones using np.ones((2, 3)).

7. How do you create an array with a range of values in NumPy?

Answer: You can create an array with a range of values using np.arange(0, 10, 2).

8. What is the shape of an array?

Answer: The shape of an array is a tuple that represents the size of the array along each dimension. You can get the shape of an array using array.shape.

9. How do you reshape an array in NumPy?

Answer: You can reshape an array using the reshape method, e.g., array.reshape((2, 3)).

10. How do you get the data type of a NumPy array?

Answer: You can get the data type of an array using array.dtype.

11. How do you create a NumPy array with random values?

Answer: You can create an array with random values using np.random.rand(2, 3).

12. How do you perform element-wise addition of two arrays?

Answer: You can perform element-wise addition using the + operator, e.g., array1 + array2.

13. How do you perform matrix multiplication in NumPy?

Answer: You can perform matrix multiplication using np.dot(array1, array2) or the @ operator.

14. How do you find the maximum value in a NumPy array?

Answer: You can find the maximum value using array.max().

15. How do you find the minimum value in a NumPy array?

Answer: You can find the minimum value using array.min().

16. How do you calculate the mean of a NumPy array?

Answer: You can calculate the mean using array.mean().

17. How do you calculate the standard deviation of a NumPy array?

Answer: You can calculate the standard deviation using array.std().

18. How do you concatenate two arrays in NumPy?

Answer: You can concatenate two arrays using np.concatenate((array1, array2), axis=0).

19. How do you stack arrays vertically in NumPy?

Answer: You can stack arrays vertically using np.vstack((array1, array2)).

20. How do you stack arrays horizontally in NumPy?

Answer: You can stack arrays horizontally using np.hstack((array1, array2)).

21. How do you get the transpose of a NumPy array?

Answer: You can get the transpose using array.T.

22. How do you save a NumPy array to a file?

Answer: You can save an array to a file using np.save('filename.npy', array).

23. How do you load a NumPy array from a file?

Answer: You can load an array from a file using np.load('filename.npy').

24. What is broadcasting in NumPy?

Answer: Broadcasting is the ability of NumPy to perform arithmetic operations on arrays of different shapes by automatically expanding the smaller array to match the shape of the larger array.

MATPLOTLIB

1. What is Matplotlib?

Answer: Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.

1. How do you install Matplotlib?

Answer: You can install Matplotlib using the pip package manager with the command `pip install matplotlib`.

2. How do you import Matplotlib in your code?

Answer: You can import Matplotlib using the command `import matplotlib.pyplot as plt`.

3. How do you create a simple line plot?

Answer:

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4])
plt.show()
```

4. How do you add labels to the x and y axes in a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.xlabel('X-axis label')
plt.ylabel('Y-axis label')
plt.show()
```

5. How do you add a title to a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.title('Plot Title')
plt.show()
```

6. How do you plot multiple lines on the same plot?

Answer:

```
plt.plot([1, 2, 3, 4], label='Line 1')
plt.plot([4, 3, 2, 1], label='Line 2')
plt.legend()
```

7. How do you create a scatter plot?

Answer:

```
plt.scatter([1, 2, 3, 4], [10, 20, 25, 30])
plt.show()
```

8. How do you create a bar plot?

Answer:

```
plt.bar([1, 2, 3, 4], [10, 20, 15, 25])
plt.show()
```

9. How do you create a histogram?

Answer:

```
data = [1, 2, 2, 3, 3, 3, 4, 4, 4, 4]
plt.hist(data, bins=4)
plt.show()
```

10. How do you change the line style in a plot?

Answer:

```
plt.plot([1, 2, 3, 4], linestyle='--')
plt.show()
```

11. How do you change the line color in a plot?

Answer:

```
plt.plot([1, 2, 3, 4], color='red')
plt.show()
```

12. How do you change the line width in a plot?

Answer:

```
plt.plot([1, 2, 3, 4], linewidth=2)
plt.show()
```

13. How do you create a subplot with multiple plots?

Answer:

```
fig, (ax1, ax2) = plt.subplots(1, 2)
ax1.plot([1, 2, 3, 4])
ax2.plot([4, 3, 2, 1])
```

14. How do you add a grid to a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.grid(True)
plt.show()
```

15. How do you save a plot to a file?

16.

Answer:

```
plt.plot([1, 2, 3, 4])
plt.savefig('plot.png')
plt.show()
```

17. How do you create a pie chart?

Answer:

```
labels = 'A', 'B', 'C', 'D'
sizes = [15, 30, 45, 10]
plt.pie(sizes, labels=labels)
plt.show()
```

18. How do you change the figure size of a plot?

Answer:

```
plt.figure(figsize=(8, 6))
plt.plot([1, 2, 3, 4])
plt.show()
```

19. How do you create a box plot?

Answer:

```
data = [np.random.normal(0, std, 100) for std in range(1, 4)]
plt.boxplot(data)
plt.show()
```

20. How do you create a heatmap?

Answer:

```
data = np.random.rand(10, 10)
plt.imshow(data, cmap='hot', interpolation='nearest')
plt.show()
```

21. How do you add annotations to a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.annotate('Important Point', xy=(2, 2), xytext=(3, 3),
arrowprops=dict(facecolor='black', shrink=0.05))
plt.show()
```

22. How do you change the font size of labels in a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.xlabel('X-axis', fontsize=14)
plt.ylabel('Y-axis', fontsize=14)
plt.show()
```

23. How do you plot a logarithmic scale?

Answer:

```
plt.plot([1, 10, 100, 1000])
plt.yscale('log')
plt.show()
```

24. How do you create an error bar plot?

Answer:

```
x = [1, 2, 3, 4]
y = [2, 3, 4, 5]
errors = [0.1, 0.2, 0.3, 0.4]
plt.errorbar(x, y, yerr=errors, fmt='o')
plt.show()
```

25. How do you create a filled area plot?

Answer:

```
x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = np.sin(x) + 0.5
plt.fill_between(x, y1, y2)
plt.show()
```

Pandas

1. What is Pandas?

Answer: Pandas is an open-source data analysis and manipulation library for Python. It provides data structures and functions needed to manipulate structured data seamlessly.

2. How do you install Pandas?

Answer: You can install Pandas using the pip package manager with the command pip install pandas.

3. How do you import Pandas in your code?

Answer: You can import Pandas using the command import pandas as pd.

4. How do you create a Pandas DataFrame from a dictionary?

Answer:

```
import pandas as pd
data = {'Name': ['Alice', 'Bob'], 'Age': [25, 30]}
df = pd.DataFrame(data)
print(df)
```

5. How do you read a CSV file into a DataFrame?

Answer:

```
import pandas as pd
df = pd.read_csv('file.csv')
print(df)
```

6. How do you write a DataFrame to a CSV file?

Answer:

```
df.to_csv('output.csv', index=False)
```

7. How do you display the first few rows of a DataFrame?

Answer:

```
print(df.head())
```

8. How do you display the last few rows of a DataFrame?

Answer:

```
print(df.tail())
```

9. How do you get the number of rows and columns in a DataFrame?

Answer:

```
print(df.shape)
```

10. How do you get a summary of a DataFrame?

Answer:

```
print(df.describe())
```

11. How do you select a column from a DataFrame?

Answer:

```
print(df['Name'])
```

12. How do you select multiple columns from a DataFrame?

Answer:

```
print(df[['Name', 'Age']])
```

13. How do you select rows based on a condition?

Answer:

```
print(df[df['Age'] > 25])
```

14. How do you add a new column to a DataFrame?

Answer:

```
df['Salary'] = [50000, 60000]
print(df)
```

15. How do you delete a column from a DataFrame?

Answer:

```
df = df.drop(columns=['Salary'])
print(df)
```

16. How do you rename columns in a DataFrame?

Answer:

```
df = df.rename(columns={'Name': 'Employee Name'})
print(df)
```

17. How do you check for missing values in a DataFrame?

Answer:

```
print(df.isnull().sum())
```

18. How do you fill missing values in a DataFrame?

Answer:

```
df['Age'] = df['Age'].fillna(0)
print(df)
```

19. How do you drop rows with missing values?

Answer:

```
df = df.dropna()
print(df)
```

20. How do you sort a DataFrame by a column?

Answer:

```
df = df.sort_values(by='Age')
print(df)
```

21. How do you group data in a DataFrame?

Answer:

```
grouped = df.groupby('Age').size()
print(grouped)
```

22. How do you apply a function to a column in a DataFrame?

Answer:

```
df['Age'] = df['Age'].apply(lambda x: x + 1)
print(df)
```

23. How do you merge two DataFrames?

Answer:

```
df1 = pd.DataFrame({'ID': [1, 2], 'Name': ['Alice', 'Bob']})
df2 = pd.DataFrame({'ID': [1, 2], 'Salary': [50000, 60000]})
merged = pd.merge(df1, df2, on='ID')
print(merged)
```

24. How do you concatenate two DataFrames?

Answer:

```
df1 = pd.DataFrame({'Name': ['Alice'], 'Age': [25]})
df2 = pd.DataFrame({'Name': ['Bob'], 'Age': [30]})
concatenated = pd.concat([df1, df2])
print(concatenated)
```

25. How do you pivot a DataFrame?

Answer:

```
df = pd.DataFrame({
    'Date': ['2021-01-01', '2021-01-02', '2021-01-01', '2021-01-02'],
    'City': ['New York', 'New York', 'Los Angeles', 'Los Angeles'],
    'Temperature': [32, 30, 75, 72]
})
pivoted = df.pivot(index='Date', columns='City', values='Temperature')
print(pivoted)
```

TensorFlow

1. What is TensorFlow?

Answer: TensorFlow is an open-source machine learning framework developed by Google that is used for building and deploying machine learning models.

2. How do you install TensorFlow?

Answer: You can install TensorFlow using the pip package manager with the command pip install tensorflow.

3. How do you import TensorFlow in your code?

Answer: You can import TensorFlow using the command import tensorflow as tf.

4. How do you create a constant tensor in TensorFlow?

Answer:

```
import tensorflow as tf
tensor = tf.constant([1, 2, 3])
print(tensor)
```

5. How do you create a variable tensor in TensorFlow?

Answer:

```
tensor = tf.Variable([1, 2, 3])
print(tensor)
```

6. How do you perform element-wise addition of two tensors?

Answer:

```
a = tf.constant([1, 2, 3])
b = tf.constant([4, 5, 6])
result = tf.add(a, b)
print(result)
```

7. How do you perform matrix multiplication in TensorFlow?

Answer:

```
a = tf.constant([[1, 2], [3, 4]])
b = tf.constant([[5, 6], [7, 8]])
result = tf.matmul(a, b)
print(result)
```

8. How do you create a tensor filled with zeros?

Answer:

```
tensor = tf.zeros([2, 3])
print(tensor)
```

9. How do you create a tensor filled with ones?

Answer:

```
tensor = tf.ones([2, 3])
print(tensor)
```

10. How do you create a tensor with a specific shape and value?

Answer:

```
tensor = tf.fill([2, 3], value=9)
print(tensor)
```

11. How do you reshape a tensor?

Answer:

```
tensor = tf.constant([[1, 2, 3], [4, 5, 6]])
reshaped = tf.reshape(tensor, [3, 2])
print(reshaped)
```

12. What is the purpose of the tf.function decorator?

Answer: The tf.function decorator converts a Python function into a TensorFlow graph for faster execution.

13. How do you create a simple sequential model in TensorFlow?

Answer:

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(64, activation='relu', input_shape=(32,)),
    Dense(10, activation='softmax')
])
```

14. How do you compile a model in TensorFlow?

Answer:

```
model.compile(optimizer='adam',  loss='sparse_categorical_crossentropy',  metrics=['accuracy'])
```

15. How do you fit a model to training data in TensorFlow?

Answer:

model.fit(x_train, y_train, epochs=10, batch_size=32)

16. How do you evaluate a model on test data in TensorFlow?

Answer:

model.evaluate(x_test, y_test)

17. How do you make predictions with a trained model in TensorFlow?

Answer:

python
Copy codepredictions = model.predict(x_new)

18. What is TensorFlow Datasets (TFDS)?

Answer: TensorFlow Datasets (TFDS) is a collection of ready-to-use datasets for use with TensorFlow, making it easier to import and work with standard datasets.

19. How do you load the MNIST dataset using TensorFlow Datasets?

Answer:

import tensorflow_datasets as tfds
mnist = tfds.load('mnist', split='train', as_supervised=True)

20. What is the purpose of tf.data API?

Answer: The tf.data API provides utilities for creating efficient input pipelines, handling large datasets, and performing preprocessing steps.

21. How do you create a dataset from a tensor in TensorFlow?

Answer:

data = tf.constant([1, 2, 3, 4, 5])
dataset = tf.data.Dataset.from_tensor_slices(data)

22. How do you batch elements of a dataset in TensorFlow?

Answer:

batched_dataset = dataset.batch(2)

23. How do you shuffle elements of a dataset in TensorFlow?

Answer:

```
shuffled_dataset = dataset.shuffle(buffer_size=10)
```

24. How do you repeat a dataset in TensorFlow?

Answer:

```
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C
o
p
y
c
o
d
e
repeated_dataset = dataset.repeat(3)
```

25. How do you save and load a TensorFlow model?

Answer:

```
# Saving the model
model.save('my_model.h5')

# Loading the model
loaded_model = tf.keras.models.load_model('my_
```